

Electrodynamics

Course notes

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I Introduction and Survey

I.1 Maxwell Equations in Vacuum, Fields, and Sources

The equations governing electromagnetic phenomena are **Maxwell's equations**:

$$\begin{aligned}\nabla \cdot \vec{D} &= \rho, \\ \nabla \times \vec{H} - \frac{\partial \vec{D}}{\partial t} &= \vec{J}, \\ \nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} &= 0, \\ \nabla \cdot \vec{B} &= 0.\end{aligned}$$

The last two equations express the absence of magnetic charges and magnetic currents. In vacuum the equations are linear (experimentally, to better than 0.1% accuracy). For external sources in vacuum, $\vec{D} = \varepsilon_0 \vec{E}$ and $\vec{B} = \mu_0 \vec{H}$, and the first two equations become

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \times \vec{B} - \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{J}, \quad c^2 = \frac{1}{\varepsilon_0 \mu_0}.$$

Notice that, taking ∂_t of $\nabla \cdot \vec{D} = \rho$ and using $\partial_t \vec{D} = \nabla \times \vec{H} - \vec{J}$,

$$\nabla \cdot \frac{\partial \vec{D}}{\partial t} = \frac{\partial \rho}{\partial t} \implies \underbrace{\nabla \cdot (\nabla \times \vec{H})}_0 - \nabla \cdot \vec{J} = \frac{\partial \rho}{\partial t},$$

which gives the **continuity equation**

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0.$$

Also essential for the motion of charged particles is the **Lorentz force** equation

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}).$$

The basic unit of charge is the magnitude of the charge of the electron,

$$|q_e| = 1.602\,177\,33(49) \times 10^{-19} \text{ C}.$$

The charges of all presently known particles or systems of particles are integral multiples of this basic unit.

I.4 Maxwell Equations in Macroscopic Media

The macroscopic Maxwell equations are such that $\vec{E}$ and $\vec{B}$ are the averages of the microscopic $\vec{E}$ and $\vec{B}$, respectively. The macroscopic field quantities $\vec{D}$ and $\vec{H}$, called the **electric displacement** and the **magnetic field** ($\vec{B}$ is called the **magnetic induction**), have components

$$\begin{aligned}D_\alpha &= \varepsilon_0 E_\alpha + \left(P_\alpha - \sum_\beta \frac{\partial Q'_{\alpha\beta}}{\partial x_\beta} + \dots \right), \\ H_\alpha &= \frac{1}{\mu_0} B_\alpha - (M_\alpha + \dots),\end{aligned}$$

where $\vec{P}$, $\vec{M}$ and $Q'_{\alpha\beta}$ are the macroscopically averaged electric dipole, magnetic dipole and electric quadrupole densities. In these equations ρ and $\vec{J}$ are the “free” sources; from the polarisation terms one can obtain the “bound” sources.

In substances other than ferroelectrics or ferromagnets, for weak enough fields the presence of an applied electric or magnetic field induces an electric or magnetic polarisation proportional to the magnitude of the applied field:

$$D_\alpha = \varepsilon_{\alpha\beta} E_\beta, \quad H_\alpha = \mu'_{\alpha\beta} B_\beta.$$

- Diamagnetic materials: no net (permanent) angular momentum.
- Paramagnetic materials: net angular momentum.

I.5 Boundary Conditions at Interfaces Between Different Media

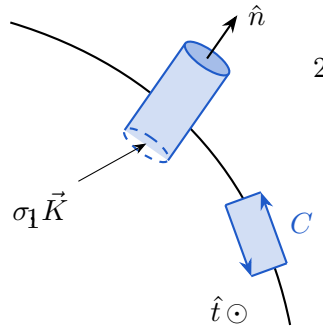


Figure 0.1: Gaussian pillbox and Stokes loop C across an interface carrying surface charge σ and surface current $\vec{K}$; $\hat{n}$ points from medium 1 to medium 2 and $\hat{t}$ is normal to the plane of the loop.

- Applying Gauss’s theorem to the pillbox, $\oint d\vec{S} \cdot \vec{D} = \int dV \rho$. With a surface charge, $\rho = \sigma \delta(s)$ and, as the pillbox height goes to zero,

$$\oint dS \hat{n} \cdot \vec{D} = \int dV \sigma \delta(s) = \int dS \sigma \implies (\vec{D}_2 - \vec{D}_1) \cdot \hat{n} = \sigma.$$

- Similarly, $\oint d\vec{S} \cdot \vec{B} = 0$ gives $(\vec{B}_2 - \vec{B}_1) \cdot \hat{n} = 0$.
- Applying Stokes’s theorem to the loop C , with $\vec{J} = \vec{K} \delta(s)$,

$$\oint_C d\vec{l} \cdot \vec{H} = \int d\vec{S} \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = \int dl \vec{K} \cdot \hat{t} \implies \hat{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}.$$

- Finally, $\oint_C d\vec{l} \cdot \vec{E} = - \int d\vec{S} \cdot \frac{\partial \vec{B}}{\partial t} \rightarrow 0$ as the loop shrinks, so $\hat{n} \times (\vec{E}_2 - \vec{E}_1) = 0$.

In summary,

$$\begin{aligned} \hat{n} \cdot (\vec{D}_2 - \vec{D}_1) &= \sigma, & \hat{n} \cdot (\vec{B}_2 - \vec{B}_1) &= 0, \\ \hat{n} \times (\vec{E}_2 - \vec{E}_1) &= 0, & \hat{n} \times (\vec{H}_2 - \vec{H}_1) &= \vec{K}. \end{aligned}$$

1 Introduction to Electrostatics

1.1 Coulomb’s Law

Experimentally, the force between two point charges at rest is

- proportional to $|q_1 q_2|$ and to $1/r^2$;
- directed along the line joining them, $\propto (\vec{r}_1 - \vec{r}_2)$;
- attractive or repulsive according to the sign of $q_1 q_2$.

The force on charge 1 due to charge 2 is

$$\vec{F}_{1/2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{\|\vec{r}_1 - \vec{r}_2\|^3} (\vec{r}_1 - \vec{r}_2).$$

1.2 Electric Field

Writing $\vec{F}_{1/2} := q_1 \vec{E}_2$, the field of a point charge q_0 at $\vec{x}_0$ is

$$\vec{E}_0(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{q_0}{\|\vec{x} - \vec{x}_0\|^3} (\vec{x} - \vec{x}_0).$$

It is experimentally observed that fields superpose:

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{\|\vec{x} - \vec{x}_i\|^3} (\vec{x} - \vec{x}_i),$$

or in general, for a charge density ρ ,

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \rho(\vec{x}') \frac{\vec{x} - \vec{x}'}{\|\vec{x} - \vec{x}'\|^3}. \quad (1.1)$$

1.3 Gauss's Law

$$\oint_{\partial V} d\vec{S} \cdot \vec{E} = \frac{1}{\epsilon_0} \int_V dV \rho.$$

1.4 Differential Form of Gauss's Law

By the divergence theorem,

$$\oint_{\partial V} d\vec{S} \cdot \vec{E} = \int_V dV \nabla \cdot \vec{E} = \frac{1}{\epsilon_0} \int_V dV \rho \implies \int_V dV \left(\nabla \cdot \vec{E} - \frac{\rho}{\epsilon_0} \right) = 0$$

for every volume V , hence

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}.$$

1.5 Another Equation of Electrostatics and the Scalar Potential

Starting from (1.1) and using $\frac{\vec{x} - \vec{x}'}{\|\vec{x} - \vec{x}'\|^3} = -\nabla \frac{1}{\|\vec{x} - \vec{x}'\|}$,

$$\vec{E}(\vec{x}) = -\frac{1}{4\pi\epsilon_0} \int d^3x' \rho(\vec{x}') \nabla \left(\frac{1}{\|\vec{x} - \vec{x}'\|} \right) = -\nabla \left(\frac{1}{4\pi\epsilon_0} \int d^3x' \frac{\rho(\vec{x}')}{\|\vec{x} - \vec{x}'\|} \right).$$

This defines the **scalar potential**

$$\phi(\vec{x}) := \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{\rho(\vec{x}')}{\|\vec{x} - \vec{x}'\|}, \quad \vec{E} = -\nabla \phi. \quad (1.2)$$

The scalar potential has a physical interpretation when we consider the work done on a test charge q in transporting it from A to B :

$$W = - \int_A^B d\vec{l} \cdot \vec{F} = -q \int_A^B d\vec{l} \cdot \vec{E} = q \int_A^B d\vec{l} \cdot \nabla \phi = q(\phi_B - \phi_A).$$

The work is independent of the path, and $q\phi$ can be interpreted as the potential energy of the test charge in the electric field. Since $\int_A^B d\vec{l} \cdot \vec{E} = -(\phi_B - \phi_A)$,

$$\oint d\vec{l} \cdot \vec{E} = 0 \quad \Rightarrow \quad \nabla \times \vec{E} = 0.$$

1.6 Surface Distributions of Charges and Dipoles and Discontinuities in the Electric Field and Potential

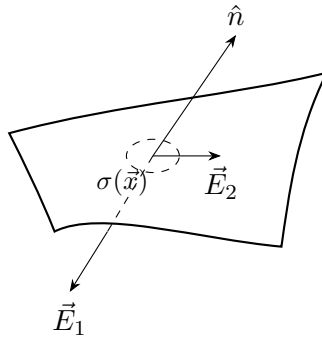


Figure 1.1: A surface S with surface charge density $\sigma(\vec{x})$.

Across a surface with charge density σ ,

$$(\vec{E}_2 - \vec{E}_1) \cdot \hat{n} = \frac{\sigma}{\epsilon_0}, \quad (\vec{E}_2 - \vec{E}_1) \times \hat{n} = 0.$$

Since $\rho d^3x = \sigma \delta(x - x_0) d^3x = \sigma d^2x$,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_S d^2x' \frac{\sigma(\vec{x}')}{\|\vec{x} - \vec{x}'\|}.$$

- For volume or surface charge distributions, the scalar potential is continuous everywhere (only the field is discontinuous across a charged surface).
- For point or line charges, or for dipole layers, the potential is no longer continuous.

Dipole layer. Consider two surfaces S and S' a distance $d(\vec{x})$ apart, carrying surface charge densities $\sigma(\vec{x})$ and $-\sigma(\vec{x})$.

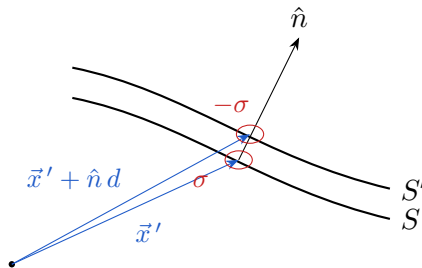


Figure 1.2: A dipole layer: two surfaces with opposite surface charge densities $\pm\sigma$ separated by $d(\vec{x})$.

The **dipole layer distribution strength** is

$$D(\vec{x}) := \lim_{d(\vec{x}) \rightarrow 0} \sigma(\vec{x}) d(\vec{x}).$$

The direction of the dipole moment of the layer is normal to the surface S and goes from the negative to the positive charge. The potential is

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_S d^2x' \frac{\sigma(\vec{x}')}{\|\vec{x} - \vec{x}'\|} - \frac{1}{4\pi\epsilon_0} \int_{S'} d^2x' \frac{\sigma(\vec{x}')}{\|\vec{x} - \vec{x}' - \hat{n} d(\vec{x}')\|},$$

and, using the expansion

$$\frac{1}{\|\vec{x} + \vec{a}\|} = \frac{1}{\|\vec{x}\|} + \vec{a} \cdot \nabla \frac{1}{\|\vec{x}\|} + \dots,$$

in the limit $d \rightarrow 0$,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_S d^2x' D(\vec{x}') \hat{n} \cdot \nabla' \left(\frac{1}{\|\vec{x} - \vec{x}'\|} \right).$$

This agrees with the potential at $\vec{x}$ caused by a dipole $\vec{p}$ at $\vec{x}'$,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot (\vec{x} - \vec{x}')}{\|\vec{x} - \vec{x}'\|^3}, \quad \vec{p} = \hat{n} D d^2x'.$$

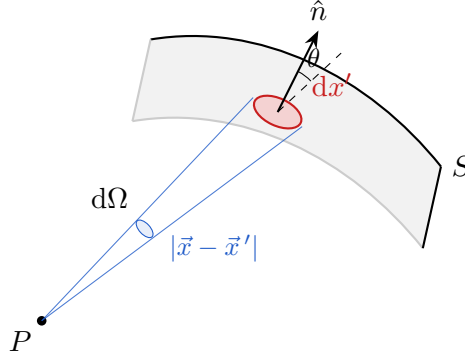


Figure 1.3: Solid angle $d\Omega$ subtended at the observation point P by the surface element $d\vec{x}'$.

Since

$$\hat{n} \cdot \nabla' \left(\frac{1}{\|\vec{x} - \vec{x}'\|} \right) d^2x' = -\frac{\cos \theta d^2x'}{\|\vec{x} - \vec{x}'\|^2} = -d\Omega,$$

the potential of a dipole layer is

$$\phi(\vec{x}) = -\frac{1}{4\pi\epsilon_0} \int_S d\Omega D(\vec{x}').$$

Just inside and just outside the layer (where the surface subtends a solid angle $\mp 2\pi$ locally), $\phi_{\text{inner}} = -D/2\epsilon_0$ and $\phi_{\text{outer}} = +D/2\epsilon_0$ (on the outer side $\cos \theta < 0$), so the potential jumps by

$$\phi_2 - \phi_1 = \frac{D}{\epsilon_0}.$$

1.7 Poisson and Laplace Equations

From

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \times \vec{E} = 0 \implies \vec{E} = -\nabla \phi,$$

we obtain the **Poisson equation**

$$\Delta \phi = -\frac{\rho}{\varepsilon_0},$$

which for $\rho = 0$ reduces to the **Laplace equation** $\Delta \phi = 0$.

The potential (1.2) is already a solution of the Poisson equation. Indeed, since

$$\begin{aligned} \Delta \left(\frac{1}{\|\vec{x} - \vec{x}'\|} \right) &= -4\pi \delta(\vec{x} - \vec{x}'), \\ \Delta \phi(\vec{x}) &= \frac{1}{4\pi\varepsilon_0} \int d^3x' \rho(\vec{x}') \Delta \left(\frac{1}{\|\vec{x} - \vec{x}'\|} \right) = -\frac{\rho(\vec{x})}{\varepsilon_0}. \end{aligned} \tag{1.3}$$

1.8 Green's Theorem

Consider the divergence theorem

$$\int_V d^3x \nabla \cdot \vec{A} = \oint_{\partial V} d^2x \hat{n} \cdot \vec{A},$$

and let $\vec{A} = \phi \nabla \psi$. Then, with $\partial\psi/\partial n := \hat{n} \cdot \nabla \psi$, we obtain **Green's first identity**

$$\int_V d^3x (\phi \Delta \psi + \nabla \phi \cdot \nabla \psi) = \oint_{\partial V} d^2x \phi \frac{\partial \psi}{\partial n}.$$

Interchanging $\phi \leftrightarrow \psi$ and subtracting, we obtain **Green's second identity** (Green's theorem)

$$\int_V d^3x (\phi \Delta \psi - \psi \Delta \phi) = \oint_{\partial V} d^2x \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right). \tag{1.4}$$

The Poisson differential equation for the potential can be converted into an integral equation by choosing $\psi = 1/R = 1/\|\vec{x} - \vec{x}'\|$, where $\vec{x}$ is the observation point and $\vec{x}'$ the integration variable. With $\Delta' \phi = -\rho/\varepsilon_0$ and (1.3),

$$\int_V d^3x' \left[-4\pi \phi(\vec{x}') \delta(\vec{x} - \vec{x}') + \frac{1}{\varepsilon_0 R} \rho(\vec{x}') \right] = \oint_{\partial V} d^2x' \left[\phi \frac{\partial}{\partial n'} \left(\frac{1}{R} \right) - \frac{1}{R} \frac{\partial \phi}{\partial n'} \right].$$

If $\vec{x}$ lies within the volume V ,

$$\phi(\vec{x}) = \frac{1}{4\pi\varepsilon_0} \int_V d^3x' \frac{\rho(\vec{x}')}{R} + \frac{1}{4\pi} \oint_{\partial V} d^2x' \left[\frac{1}{R} \frac{\partial \phi}{\partial n'} - \phi \frac{\partial}{\partial n'} \left(\frac{1}{R} \right) \right].$$

1.9 Uniqueness of the Solution with Dirichlet or Neumann Boundary Conditions

There are two types of boundary conditions:

- **Dirichlet:** $\phi|_{\partial V}$ is known;
- **Neumann:** $\partial\phi/\partial n|_{\partial V}$ is known.

Either one guarantees the existence of the solution. In general there is no solution if both $\phi|_{\partial V}$ and $\partial\phi/\partial n|_{\partial V}$ are specified arbitrarily on a closed surface (Cauchy boundary conditions): the problem is overdetermined.

Uniqueness. Consider Green's first identity with $\phi = \psi = U := \phi_2 - \phi_1$, where ϕ_1 and ϕ_2 are solutions of the Poisson equation with the same boundary conditions. Then $\Delta U = 0$ in V and either $U = 0$ or $\partial U / \partial n = 0$ on ∂V , so

$$\int_V d^3x \left(U \underbrace{\Delta U}_0 + \nabla U \cdot \nabla U \right) = \oint_{\partial V} d^2x \underbrace{U \frac{\partial U}{\partial n}}_0 \implies \int_V d^3x |\nabla U|^2 = 0 \implies \nabla U|_V = 0,$$

i.e. U is constant in V .

- Dirichlet: $U|_{\partial V} = 0 \Rightarrow U|_V = 0 \Rightarrow \phi_1 = \phi_2$.
- Neumann: $\partial U / \partial n|_{\partial V} = 0 \Rightarrow U|_V = \text{const} \Rightarrow \phi_1 = \phi_2 + \text{const}$.

1.10 Formal Solution of the Electrostatic Boundary-Value Problem with Green Function

So far we have used $\Delta'(1/\|\vec{x} - \vec{x}'\|) = -4\pi\delta(\vec{x} - \vec{x}')$. However, $1/\|\vec{x} - \vec{x}'\|$ is only one of a class of functions, called **Green functions**, satisfying

$$\Delta'G(\vec{x}, \vec{x}') = -4\pi\delta(\vec{x} - \vec{x}'),$$

where

$$G(\vec{x}, \vec{x}') = \frac{1}{\|\vec{x} - \vec{x}'\|} + F(\vec{x}, \vec{x}'), \quad \Delta'F(\vec{x}, \vec{x}') = 0.$$

The function F can be chosen so that one of the boundary terms is eliminated (this is the idea behind the **method of images**). For Dirichlet boundary conditions G is symmetric, $G(\vec{x}, \vec{x}') = G(\vec{x}', \vec{x})$: source and observation points can be interchanged.

Using $\phi = \phi$ and $\psi = G(\vec{x}, \vec{x}')$ in Green's second identity (1.4),

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V d^3x' \rho(\vec{x}') G(\vec{x}, \vec{x}') + \frac{1}{4\pi} \oint_{\partial V} d^2x' \left[G(\vec{x}, \vec{x}') \frac{\partial \phi}{\partial n'} - \phi \frac{\partial G}{\partial n'} \right].$$

Dirichlet boundary conditions. Choosing $G_D(\vec{x}, \vec{x}')|_{\partial V} = 0$,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V d^3x' \rho(\vec{x}') G_D(\vec{x}, \vec{x}') - \frac{1}{4\pi} \oint_{\partial V} d^2x' \phi(\vec{x}') \frac{\partial G_D}{\partial n'}.$$

Neumann boundary conditions. One must be careful, since $\partial G / \partial n'|_{\partial V} \neq 0$ in general. In fact, integrating $\Delta'G(\vec{x}, \vec{x}') = -4\pi\delta(\vec{x} - \vec{x}')$ over V ,

$$\int_V d^3x' \nabla' \cdot (\nabla' G) = -4\pi \implies \oint_{\partial V} d^2x' \frac{\partial G}{\partial n'} = -4\pi.$$

Therefore we should impose

$$\left. \frac{\partial G_N}{\partial n'} \right|_{\partial V} = -\frac{4\pi}{S},$$

where S is the area of the boundary surface. Then

$$\phi(\vec{x}) = \langle \phi \rangle_{\partial V} + \frac{1}{4\pi\epsilon_0} \int_V d^3x' \rho(\vec{x}') G_N(\vec{x}, \vec{x}') + \frac{1}{4\pi} \oint_{\partial V} d^2x' G_N(\vec{x}, \vec{x}') \frac{\partial \phi}{\partial n'},$$

where $\langle \phi \rangle_{\partial V}$ is the average of ϕ over the boundary.

1.11 Electrostatic Potential Energy and Energy Density; Capacitance

If a point charge q_i is brought from infinity to a point $\vec{x}_i$ in a region of localised electric field, the work done on the charge is

$$W_i = q_i \phi(\vec{x}_i).$$

If the potential is produced by an array of $n - 1$ charges,

$$\phi(\vec{x}_i) = \frac{1}{4\pi\epsilon_0} \sum_{j=1}^{n-1} \frac{q_j}{|\vec{x}_i - \vec{x}_j|},$$

the potential energy of the charge q_i is

$$W_i = \frac{1}{4\pi\epsilon_0} \sum_{j=1}^{n-1} \frac{q_i q_j}{|\vec{x}_i - \vec{x}_j|}.$$

The total potential energy of all the charges due to all the forces acting between them is

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j < i} \frac{q_i q_j}{|\vec{x}_i - \vec{x}_j|} = \frac{1}{8\pi\epsilon_0} \sum_{\substack{i,j \\ i \neq j}} \frac{q_i q_j}{|\vec{x}_i - \vec{x}_j|}$$

(the terms $i = j$ would be infinite self-energy terms). For continuous charge distributions,

$$W = \frac{1}{8\pi\epsilon_0} \int d^3x d^3x' \frac{\rho(\vec{x})\rho(\vec{x}')}{|\vec{x} - \vec{x}'|},$$

which is equivalent to

$$W = \frac{1}{2} \int d^3x \rho(\vec{x}) \phi(\vec{x}).$$

Using $\rho = -\epsilon_0 \Delta\phi$ and integrating by parts,

$$W = -\frac{\epsilon_0}{2} \int d^3x \phi \Delta\phi = \frac{\epsilon_0}{2} \int d^3x |\nabla\phi|^2,$$

and, with $\vec{E} = -\nabla\phi$,

$$W = \int d^3x \frac{\epsilon_0}{2} |\vec{E}|^2.$$

The quantity $w := \frac{\epsilon_0}{2} E^2$ is the **energy density**. It is always positive, because it contains the self-energy contributions.

Systems of conductors. For a system of n conductors, the electrostatic potential energy can be expressed in terms of the potentials and charges alone. The potentials are linear in the charges,

$$V_i = \sum_j p_{ij} Q_j,$$

or, inverting,

$$Q_i = \sum_j C_{ij} V_j,$$

where the C_{ii} are the **capacities** (capacitances) and the C_{ij} , $i \neq j$, the **coefficients of induction**. The capacitance of a conductor is the total charge on the conductor when it is maintained at unit potential, all other conductors being held at zero potential. The potential energy of the system of conductors is

$$W = \frac{1}{2} \sum_i Q_i V_i = \frac{1}{2} \sum_{i,j} C_{ij} V_i V_j.$$

1.12 Variational Approach to the Solution of the Laplace and Poisson Equations

The far-reaching concept that physical systems in equilibrium have minimal energy content is generalised to the consideration of energy-like functionals.

Dirichlet problem. Consider the functional

$$I[\psi] = \int_V d^3x \left(\frac{1}{2} |\nabla\psi|^2 - g\psi \right).$$

Its variation, $\delta I := I[\psi + \delta\psi] - I[\psi]$, is, to first order,

$$\delta I = \int_V d^3x (-\Delta\psi - g) \delta\psi + \oint_{\partial V} d^2x \delta\psi \frac{\partial\psi}{\partial n},$$

where the surface term vanishes for Dirichlet boundary conditions ($\delta\psi = 0$ on ∂V). Stationarity gives $\Delta\psi = -g$, which for $\psi = \phi$ and $g = \rho/\varepsilon_0$ is the Poisson equation $\Delta\phi = -\rho/\varepsilon_0$.

Neumann problem. The appropriate functional is

$$I[\psi] = \int_V d^3x \left(\frac{1}{2} |\nabla\psi|^2 - g\psi \right) - \oint_{\partial V} d^2x f\psi, \quad f = \frac{\partial\psi}{\partial n} \Big|_{\partial V}.$$

The stationary nature of the extremum of $I[\psi]$ permits a practical approach to an approximate solution for $\psi(\vec{x})$:

1. choose a trial function $\psi(\vec{x}) = A \Psi(\vec{x}; \alpha, \beta, \gamma, \dots)$ that satisfies the boundary conditions, where A is a normalisation constant and $\alpha, \beta, \gamma, \dots$ are parameters;
2. substitute it in $I[\psi]$ and vary the parameters to find the minimum.

2 Boundary-Value Problems in Electrostatics: I

2.1 Method of Images

The method of images concerns itself with the problem of one or more point charges in the presence of boundary surfaces, for example conductors either grounded or held at fixed potentials. Under favourable conditions it is possible to infer from the geometry of the situation that a small number of suitably placed charges of appropriate magnitudes, **external to the region of interest**, can simulate the required boundary conditions.

2.2 Point Charge in the Presence of a Grounded Conducting Sphere

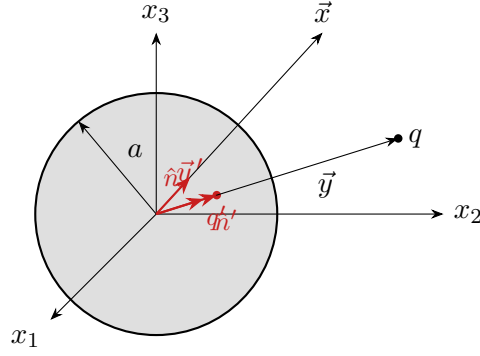


Figure 2.1: Point charge q at $\vec{y}$ outside a grounded conducting sphere of radius a , and its image q' at $\vec{y}'$.

We look for the potential as that of q plus an image charge q' inside the sphere:

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{x} - \vec{y}|} + \frac{1}{4\pi\epsilon_0} \frac{q'}{|\vec{x} - \vec{y}'|}.$$

Writing $\vec{x} = x \hat{n}$, $\vec{y} = y \hat{n}'$ and $\vec{y}' = y' \hat{n}'$,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|x\hat{n} - y\hat{n}'|} + \frac{q'}{|x\hat{n} - y'\hat{n}'|} \right],$$

and the boundary condition $\phi(x = a) = 0$ reads

$$0 = \frac{q}{a |\hat{n} - \frac{y}{a} \hat{n}'|} + \frac{q'}{y' |\hat{n}' - \frac{a}{y'} \hat{n}|}.$$

For this to hold for all possible values of $\hat{n} \cdot \hat{n}'$ we need

$$\frac{q}{a} = -\frac{q'}{y'}, \quad \frac{y}{a} = \frac{a}{y'} \implies y' = \frac{a^2}{y}, \quad q' = -q \frac{a}{y}.$$

Then, for any $|\vec{x}| > a$, with $\hat{n} \cdot \hat{n}' = \cos \gamma$,

$$\begin{aligned} \phi(\vec{x}) &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\vec{x} - \vec{y}|} - \frac{qa}{y |\vec{x} - \frac{a^2}{y^2} \vec{y}|} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{x^2 + y^2 - 2xy \cos \gamma}} - \frac{a}{y \sqrt{x^2 + \frac{a^4}{y^2} - 2x \frac{a^2}{y} \cos \gamma}} \right]. \end{aligned}$$

Using Gauss's law at the boundary, with $\vec{E}_{\text{in}} = 0$ inside the conductor, $E|_{\partial V} = \sigma/\epsilon_0$, i.e. $-\partial\phi/\partial x|_{\partial V} = \sigma/\epsilon_0$, so the induced surface charge density is

$$\sigma = -\epsilon_0 \left. \frac{\partial\phi}{\partial x} \right|_{x=a}.$$

The force on q can be computed as the force on q due to the image charge q' .

2.5 Conducting Sphere in a Uniform Electric Field by the Method of Images

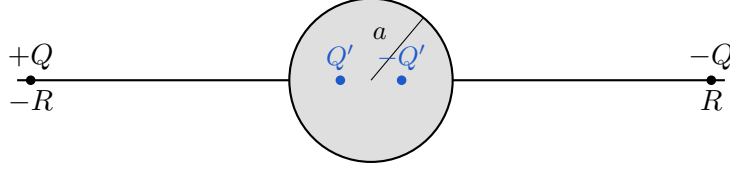


Figure 2.2: Two charges $\pm Q$ at $z = \mp R$ and their images inside the sphere. For $R \rightarrow \infty$ with Q/R^2 fixed they produce a uniform field near the sphere.

If $R \rightarrow \infty$ ($R \gg a$) while Q/R^2 is kept finite, the electric field in the neighbourhood of the conductor is constant, with magnitude $E_0 = 2Q/(4\pi\epsilon_0 R^2)$. Using the images of Section 2.2 (charges $\mp Qa/R$ at $z = \mp a^2/R$), the potential is

$$\phi(\vec{r}) = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{r^2 + R^2 + 2rR\cos\theta}} - \frac{1}{\sqrt{r^2 + R^2 - 2rR\cos\theta}} \right. \\ \left. - \frac{a}{R\sqrt{r^2 + \frac{a^4}{R^2} + \frac{2a^2r}{R}\cos\theta}} + \frac{a}{R\sqrt{r^2 + \frac{a^4}{R^2} - \frac{2a^2r}{R}\cos\theta}} \right].$$

For $R \rightarrow \infty$,

$$\frac{1}{\sqrt{r^2 + R^2 \pm 2rR\cos\theta}} \simeq \frac{1}{R} \left(1 \mp \frac{r}{R} \cos\theta \right), \quad \frac{a}{R\sqrt{r^2 + \frac{a^4}{R^2} \pm \frac{2a^2r}{R}\cos\theta}} \simeq \frac{a}{Rr} \left(1 \mp \frac{a^2}{Rr} \cos\theta \right),$$

up to terms that vanish in the limit, so that

$$\phi(\vec{r}) = \frac{Q}{4\pi\epsilon_0} \left[-\frac{2r}{R^2} \cos\theta + \frac{2a^3}{r^2 R^2} \cos\theta \right],$$

that is,

$$\phi(\vec{r}) = -E_0 \left(r - \frac{a^3}{r^2} \right) \cos\theta.$$

- The first term, $-E_0 r \cos\theta = -E_0 z$, is the potential of a uniform electric field.
- The second term is the potential due to the induced surface-charge density: the image charges form a dipole of strength $D = \frac{Qa}{R} \cdot \frac{2a^2}{R} = 4\pi\epsilon_0 E_0 a^3$.

The induced surface-charge density is

$$\sigma = -\epsilon_0 \frac{\partial\phi}{\partial r} \Big|_{r=a} = 3\epsilon_0 E_0 \cos\theta, \quad \text{so} \quad \phi = -\frac{\sigma(\theta)}{3\epsilon_0} \left(r - \frac{a^3}{r^2} \right).$$

2.6 Green Function for the Sphere; General Solution for the Potential

In Section 2.2 we computed the potential in a region where there was just a point source of charge, $\rho = q\delta(\vec{x} - \vec{x}')$. That is, we solved

$$\Delta\phi = -\frac{q}{\epsilon_0} \delta(\vec{x} - \vec{x}').$$

Notice that if we replace $q \rightarrow 4\pi\epsilon_0$, this becomes $\Delta G(\vec{x}, \vec{x}') = -4\pi\delta(\vec{x} - \vec{x}')$. Hence the Green function for a grounded conducting sphere is

$$G(\vec{x}, \vec{x}') = \frac{1}{|\vec{x} - \vec{x}'|} - \frac{a}{x' |\vec{x} - \frac{a^2}{x'^2} \vec{x}'|}.$$

Thus, in general, if the boundary condition is $\phi|_{r=a} = \phi(a, \theta, \varphi)$ and there is no charge outside the sphere, the solution is

$$\phi(\vec{x}) = -\frac{1}{4\pi} \oint_{\partial V} d^2x' \phi(a, \theta', \varphi') \frac{\partial G}{\partial n'} \Big|_{x'=a}$$

(plus $\frac{1}{4\pi\epsilon_0} \int_V d^3x' \rho(\vec{x}') G(\vec{x}, \vec{x}')$ if there are charges in V).

2.8 Orthogonal Functions and Expansions

The particular orthogonal set chosen depends on the symmetries or near symmetries involved. General properties of orthogonal functions and of expansions in terms of them:

Consider an interval (a, b) in a variable ξ with a set of real or complex functions $U_n(\xi)$, $n = 1, 2, \dots$, square integrable and orthogonal on (a, b) .

- **Orthonormality:**

$$\int_a^b d\xi U_n^*(\xi) U_m(\xi) = \delta_{nm}.$$

- **Completeness** is defined by the requirement that, for every arbitrarily small number, there exists a finite N_0 such that for $N > N_0$ the mean square error

$$M_N = \int_a^b d\xi \left| f(\xi) - \sum_{n=1}^N a_n U_n(\xi) \right|^2$$

is smaller than that number. Then the series representation $\sum_{n=1}^\infty a_n U_n(\xi) = f(\xi)$ is said to **converge in the mean** to $f(\xi)$.

Notice that if $f(\xi) = \sum_n a_n U_n(\xi)$, multiplying by $U_m^*(\xi)$ and integrating,

$$\int_a^b d\xi f(\xi) U_m^*(\xi) = \sum_n a_n \underbrace{\int_a^b d\xi U_m^*(\xi) U_n(\xi)}_{\delta_{mn}} \implies a_n = \int_a^b d\xi f(\xi) U_n^*(\xi).$$

Moreover,

$$f(\xi) = \sum_n a_n U_n(\xi) = \sum_n \int_a^b d\xi' f(\xi') U_n^*(\xi') U_n(\xi) = \int_a^b d\xi' f(\xi') \left(\sum_n U_n^*(\xi') U_n(\xi) \right),$$

which gives the **completeness** or **closure relation**

$$\sum_n U_n^*(\xi') U_n(\xi) = \delta(\xi - \xi').$$

Examples of orthonormal sets.

- On $x \in (-a/2, a/2)$: $\sqrt{\frac{2}{a}} \sin\left(\frac{2\pi n x}{a}\right)$, $\sqrt{\frac{2}{a}} \cos\left(\frac{2\pi n x}{a}\right)$, $n \in \mathbb{N}$, together with $1/\sqrt{a}$ for $n = 0$. This gives the Fourier series

$$f(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{2\pi n x}{a}\right) + B_n \sin\left(\frac{2\pi n x}{a}\right) \right],$$

$$A_n = \frac{2}{a} \int_{-a/2}^{a/2} dx f(x) \cos\left(\frac{2\pi n x}{a}\right), \quad B_n = \frac{2}{a} \int_{-a/2}^{a/2} dx f(x) \sin\left(\frac{2\pi n x}{a}\right).$$

- $U_n(x) = \frac{1}{\sqrt{a}} e^{i2\pi nx/a}$, $n = 0, \pm 1, \pm 2, \dots$:

$$f(x) = \frac{1}{\sqrt{a}} \sum_{n=-\infty}^{\infty} A_n e^{i2\pi nx/a}, \quad A_n = \frac{1}{\sqrt{a}} \int_{-a/2}^{a/2} dx f(x) e^{-i2\pi nx/a}.$$

If the interval becomes infinite ($a \rightarrow \infty$),

$$\frac{2\pi n}{a} \rightarrow k, \quad \sum_n \rightarrow \int_{-\infty}^{\infty} dn = \frac{a}{2\pi} \int_{-\infty}^{\infty} dk, \quad A_n \rightarrow \sqrt{\frac{2\pi}{a}} A(k),$$

and we obtain the Fourier integral

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk A(k) e^{ikx}, \quad A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x) e^{-ikx},$$

with

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} dx e^{i(k-k')x} &= \delta(k - k') && \text{(orthogonality),} \\ \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{ik(x-x')} &= \delta(x - x') && \text{(completeness).} \end{aligned}$$

2.9 Separation of Variables; Laplace Equation in Rectangular Coordinates

The Laplace equation in rectangular coordinates is

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0.$$

With $\phi(x, y, z) = X(x)Y(y)Z(z)$,

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0 \implies \frac{1}{X} \frac{d^2 X}{dx^2} = -\alpha^2, \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = -\beta^2, \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = \gamma^2,$$

where $\alpha^2 + \beta^2 = \gamma^2$. To determine α and β it is necessary to impose the boundary conditions.

3 Boundary-Value Problems in Electrostatics: II

3.1 Laplace Equation in Spherical Coordinates

In spherical coordinates (r, θ, φ) the Laplace equation can be written in the form

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r\Phi) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \Phi}{\partial \varphi^2} = 0.$$

Using separation of variables, $\Phi = \frac{U(r)}{r} P(\theta) Q(\varphi)$,

$$r^2 \sin^2 \theta \left[\frac{1}{U} \frac{d^2 U}{dr^2} + \frac{1}{Pr^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) \right] + \frac{1}{Q} \frac{d^2 Q}{d\varphi^2} = 0,$$

and therefore

$$\begin{aligned} \frac{1}{Q} \frac{d^2 Q}{d\varphi^2} &= -m^2, \\ \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) + \left[l(l+1) - \frac{m^2}{\sin^2 \theta} \right] P &= 0, \\ \frac{d^2 U}{dr^2} - \frac{l(l+1)}{r^2} U &= 0, \end{aligned}$$

with solutions

$$Q = e^{\pm im\varphi}, \quad m \in \mathbb{N}_0, \quad U = A r^{l+1} + B r^{-l}.$$

(m must be an integer for Q to be single-valued.)

3.2 Legendre Equation and Legendre Polynomials

The θ equation for $P(\theta)$ is customarily expressed in terms of $x = \cos \theta$ as the **generalised Legendre equation**

$$\frac{d}{dx} \left[(1-x^2) \frac{dP}{dx} \right] + \left[l(l+1) - \frac{m^2}{1-x^2} \right] P = 0.$$

For $m = 0$ this is the **Legendre differential equation**

$$\frac{d}{dx} \left[(1-x^2) \frac{dP}{dx} \right] + l(l+1)P = 0, \quad -1 \leq x \leq 1.$$

Taking the ansatz $P(x) = x^\alpha \sum_{j=0}^{\infty} a_j x^j$,

$$\sum_{j=0}^{\infty} \left\{ (\alpha+j)(\alpha+j-1) a_j x^{\alpha+j-2} - [(\alpha+j)(\alpha+j+1) - l(l+1)] a_j x^{\alpha+j} \right\} = 0.$$

The lowest powers require

$$a_0 \neq 0 : \alpha(\alpha-1) = 0, \quad a_1 \neq 0 : \alpha(\alpha+1) = 0,$$

which are equivalent choices ($a_0 \neq 0$ or $a_1 \neq 0$, but not both), and the recursion relation is

$$a_{j+2} = \left[\frac{(\alpha+j)(\alpha+j+1) - l(l+1)}{(\alpha+j+1)(\alpha+j+2)} \right] a_j.$$

The series converges for $x^2 < 1$, regardless of l , but diverges at $x = \pm 1$ unless it terminates. It terminates only if $l \in \mathbb{N}_0$ (with the appropriate choice of α), and then gives polynomials.

By convention the polynomials are normalised to have value unity at $x = +1$, and are called **Legendre polynomials** of order l , $P_l(x)$. They are given by **Rodrigues' formula**

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l.$$

The Legendre polynomials form a complete set of orthogonal functions on $[-1, 1]$:

$$\int_{-1}^1 dx P_{l'}(x) P_l(x) = \frac{2}{2l+1} \delta_{ll'},$$

$$f(x) = \sum_{l=0}^{\infty} A_l P_l(x), \quad -1 \leq x \leq 1, \quad A_l = \frac{2l+1}{2} \int_{-1}^1 dx f(x) P_l(x).$$

3.3 Boundary-Value Problems with Azimuthal Symmetry

If there is azimuthal symmetry, $m = 0$, and

$$\Phi(r, \theta) = \sum_l (A_l r^l + B_l r^{-(l+1)}) P_l(\cos \theta),$$

where the coefficients A_l, B_l depend on the boundary conditions.

An important result is

$$\frac{1}{|\vec{x} - \vec{x}'|} = \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \gamma), \quad (3.1)$$

where $r_{<} (r_{>})$ is the smaller (larger) of $|\vec{x}|$ and $|\vec{x}'|$, and γ is the angle between $\vec{x}$ and $\vec{x}'$. The left-hand side satisfies the Laplace equation (for $\vec{x} \neq \vec{x}'$), and if $\vec{x}$ is on the z axis through $\vec{x}'$ the expansion reduces to $\sum_l r_{<}^l / r_{>}^{l+1}$.

In summary, if $\phi(r, \theta = 0)$ is known on the axis and expressed as $\sum_l a_l(r)$, with $a_l(r) = A_l r^l + B_l r^{-(l+1)}$, then the general solution off the z axis is given by

$$\phi(r, \theta) = \sum_l a_l(r) P_l(\cos \theta).$$

3.5 Associated Legendre Functions and the Spherical Harmonics $Y_{lm}(\theta, \varphi)$

For problems with azimuthal symmetry ($m = 0$) and $\theta \in [0, \pi]$ the angular solution involves only the Legendre polynomials. In essentially the same manner as for the ordinary Legendre functions, it can be shown that to have finite solutions on the interval $-1 \leq x \leq 1$ the parameter l must satisfy $l \in \mathbb{N}_0$ and the integer $m \in \{-l, -l+1, \dots, l-1, l\}$. The solution with these properties is called an **associated Legendre function** $P_l^m(x)$. For $m > 0$,

$$P_l^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_l(x),$$

and, using Rodrigues' formula,

$$P_l^m(x) = \frac{(-1)^m}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} (x^2-1)^l,$$

which also holds for negative m , with

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x).$$

For a fixed m the associated Legendre functions form an orthogonal set:

$$\int_{-1}^1 dx P_{l'}^m(x) P_l^m(x) = \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \delta_{l'l}.$$

The solution of the Laplace equation was decomposed into a product of factors for the three variables r, θ, φ . It is convenient to combine the angular factors and construct orthonormal functions over the unit sphere, called **spherical harmonics**:

$$Y_{lm}(\theta, \varphi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\varphi}, \quad Y_{l,-m}(\theta, \varphi) = (-1)^m Y_{lm}^*(\theta, \varphi).$$

The normalisation and orthogonality conditions are

$$\int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta Y_{l'm'}^*(\theta, \varphi) Y_{lm}(\theta, \varphi) = \delta_{l'l} \delta_{m'm},$$

and the completeness relation is

$$\sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}^*(\theta', \varphi') Y_{lm}(\theta, \varphi) = \delta(\varphi - \varphi') \delta(\cos \theta - \cos \theta').$$

An arbitrary function $g(\theta, \varphi)$ can be expanded in spherical harmonics:

$$g(\theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} Y_{lm}(\theta, \varphi), \quad A_{lm} = \int d\Omega Y_{lm}^*(\theta, \varphi) g(\theta, \varphi).$$

The general solution of a boundary-value problem in spherical coordinates can be written in terms of spherical harmonics and powers of r :

$$\phi(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l [A_{lm} r^l + B_{lm} r^{-(l+1)}] Y_{lm}(\theta, \varphi).$$

3.6 Addition Theorem for Spherical Harmonics

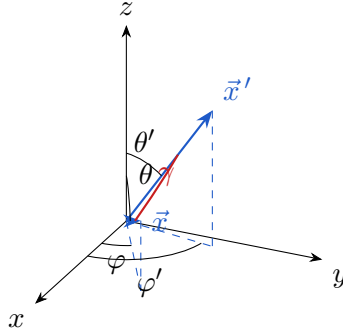


Figure 3.1: Angles of $\vec{x}$ and $\vec{x}'$, and the angle γ between them.

With $\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\varphi - \varphi')$, the **addition theorem** states

$$P_l(\cos \gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}^*(\theta', \varphi') Y_{lm}(\theta, \varphi).$$

Notice that for $\gamma = 0$,

$$\sum_{m=-l}^l |Y_{lm}(\theta, \varphi)|^2 = \frac{2l+1}{4\pi}.$$

The addition theorem can be used in (3.1) to expand $1/|\vec{x} - \vec{x}'|$ ($\vec{x}$ the field point, $\vec{x}'$ the location of the charge) into its most explicit form,

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\theta', \varphi') Y_{lm}(\theta, \varphi), \quad (3.2)$$

which is completely factorised in the coordinates $\vec{x}$ and $\vec{x}'$, and therefore useful in integrations.

3.9 Expansion of Green Functions in Spherical Coordinates

A Green function for a Dirichlet potential problem satisfies

$$\Delta G(\vec{x}, \vec{x}') = -4\pi \delta(\vec{x} - \vec{x}'), \quad G(\vec{x}, \vec{x}')|_{\partial V} = 0.$$

Recall that

$$\delta(\vec{x} - \vec{x}') = \frac{1}{r^2} \delta(r - r') \delta(\varphi - \varphi') \delta(\cos \theta - \cos \theta') = \frac{1}{r^2} \delta(r - r') \sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}^*(\theta', \varphi') Y_{lm}(\theta, \varphi).$$

Using the ansatz

$$G(\vec{x}, \vec{x}') = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm}(r; r', \theta', \varphi') Y_{lm}(\theta, \varphi),$$

we find $A_{lm}(r; r', \theta', \varphi') = g_l(r, r') Y_{lm}^*(\theta', \varphi')$, with

$$\frac{1}{r} \frac{d^2}{dr^2} (r g_l(r, r')) - \frac{l(l+1)}{r^2} g_l(r, r') = -\frac{4\pi}{r^2} \delta(r - r'),$$

so that

$$g_l(r, r') = \begin{cases} A r^l + B r^{-(l+1)}, & r < r', \\ A' r^l + B' r^{-(l+1)}, & r > r'. \end{cases}$$

Assuming the boundary conditions $\phi|_{r=a} = \phi|_{r=b} = 0$ (a spherical shell $a < r < b$),

$$g_l(r, r') = \begin{cases} A \left(r^l - \frac{a^{2l+1}}{r^{l+1}} \right), & r < r', \\ B' \left(\frac{1}{r^{l+1}} - \frac{r^l}{b^{2l+1}} \right), & r > r'. \end{cases}$$

By the symmetry $g_l(r, r') = g_l(r', r)$,

$$g_l(r, r') = C \left(r_{<}^l - \frac{a^{2l+1}}{r_{<}^{l+1}} \right) \left(\frac{1}{r_{>}^{l+1}} - \frac{r_{>}^l}{b^{2l+1}} \right),$$

where $r_{>}$ ($r_{<}$) is the larger (smaller) of r and r' . The constant C is determined by integrating the equation defining the Green function across $r = r'$, which takes into account the delta function:

$$C = \frac{4\pi}{(2l+1) \left[1 - \left(\frac{a}{b} \right)^{2l+1} \right]}.$$

Therefore the Green function for a spherical shell bounded by $r = a$ and $r = b$ is

$$G(\vec{x}, \vec{x}') = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{Y_{lm}^*(\theta', \varphi') Y_{lm}(\theta, \varphi)}{(2l+1) \left[1 - \left(\frac{a}{b} \right)^{2l+1} \right]} \left(r_{<}^l - \frac{a^{2l+1}}{r_{<}^{l+1}} \right) \left(\frac{1}{r_{>}^{l+1}} - \frac{r_{>}^l}{b^{2l+1}} \right).$$

4 Multipoles, Electrostatics of Macroscopic Media, Dielectrics

4.1 Multipole Expansion

Consider the potential of a localised charge distribution $\rho(\vec{x})$,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|}.$$

Using (3.2) for $r = |\vec{x}|$ outside the distribution,

$$\phi(\vec{x}) = \frac{1}{\epsilon_0} \sum_{l,m} \frac{1}{2l+1} \left[\int d^3x' Y_{lm}^*(\theta', \varphi') r'^l \rho(\vec{x}') \right] \frac{Y_{lm}(\theta, \varphi)}{r^{l+1}}.$$

The **multipole moments** are

$$q_{lm} := \int d^3x' Y_{lm}^*(\theta', \varphi') r'^l \rho(\vec{x}'), \quad q_{l,-m} = (-1)^m q_{lm}^*,$$

and

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \sum_{l,m} \frac{4\pi}{2l+1} q_{lm} \frac{Y_{lm}(\theta, \varphi)}{r^{l+1}}.$$

The l -th multipole coefficients ($2l+1$ in number) are linear combinations of the corresponding multipoles expressed in rectangular coordinates:

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r} + \frac{\vec{p} \cdot \vec{x}}{r^3} + \frac{1}{2} \sum_{i,j} Q_{ij} \frac{x_i x_j}{r^5} + \dots \right], \quad Q_{ij} := \int d^3x' (3x'_i x'_j - r'^2 \delta_{ij}) \rho(\vec{x}').$$

The electric field of a given multipole (l, m) is most easily expressed in spherical coordinates:

$$\begin{aligned} E_r &= \frac{l+1}{(2l+1)\epsilon_0} q_{lm} \frac{Y_{lm}(\theta, \varphi)}{r^{l+2}}, \\ E_\theta &= -\frac{1}{(2l+1)\epsilon_0} q_{lm} \frac{1}{r^{l+2}} \frac{\partial}{\partial \theta} Y_{lm}(\theta, \varphi), \\ E_\varphi &= -\frac{1}{(2l+1)\epsilon_0} q_{lm} \frac{1}{r^{l+2}} \frac{\partial}{\partial \varphi} Y_{lm}(\theta, \varphi). \end{aligned}$$

4.2 Multipole Expansion of the Energy of a Charge Distribution in an External Field

If a localised charge distribution $\rho(\vec{x})$ is placed in an external potential $\phi(\vec{x})$, the electrostatic energy of the system is

$$W = \int d^3x \rho(\vec{x}) \phi(\vec{x}).$$

If the potential is slowly varying over the region where $\rho(\vec{x})$ is non-negligible,

$$\begin{aligned} \phi(\vec{x}) &= \phi(0) + \vec{x} \cdot \nabla \phi(0) + \frac{1}{2} \sum_{i,j} x_i x_j \frac{\partial^2 \phi}{\partial x_i \partial x_j} + \dots \\ &= \phi(0) - \vec{x} \cdot \vec{E}(0) - \frac{1}{2} \sum_{i,j} x_i x_j \frac{\partial E_j}{\partial x_i} + \dots \end{aligned}$$

Since $\nabla \cdot \vec{E} = 0$ for the external field, we can subtract $\frac{1}{6} r^2 \nabla \cdot \vec{E}(0) = 0$:

$$\phi(\vec{x}) = \phi(0) - \vec{x} \cdot \vec{E}(0) - \frac{1}{6} \sum_{i,j} (3x_i x_j - r^2 \delta_{ij}) \frac{\partial E_j}{\partial x_i} + \dots,$$

and therefore

$$W = q \phi(0) - \vec{p} \cdot \vec{E}(0) - \frac{1}{6} \sum_{i,j} Q_{ij} \frac{\partial E_j}{\partial x_i} + \dots$$

The charge interacts with the potential, the dipole moment with the electric field, and the quadrupole moment with the field gradient.

4.3 Elementary Treatment of Electrostatics with Ponderable Media

We go from the microscopic to the macroscopic description. Notice that $\nabla \times \vec{E}_{\text{micro}} = 0$ remains true after averaging:

$$\nabla \times \vec{E} = 0.$$

On the other hand, if an electric field is applied to a dielectric object, an electric polarisation is produced,

$$\vec{P}(\vec{x}) = \sum_i N_i \langle \vec{p}_i \rangle,$$

where $\langle \vec{p}_i \rangle$ is the average dipole moment of a molecule of type i and N_i the average number of such molecules per unit volume. Moreover, the charge density is

$$\rho(\vec{x}) = \sum_i N_i \langle e_i \rangle + \rho_{\text{free}}.$$

If no higher multipoles are present,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \left[\frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} + \frac{\vec{P}(\vec{x}') \cdot (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \right].$$

Integrating the second term by parts,

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{|\vec{x} - \vec{x}'|} [\rho(\vec{x}') - \nabla' \cdot \vec{P}(\vec{x}')],$$

and hence

$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0} [\rho(\vec{x}) - \nabla \cdot \vec{P}(\vec{x})].$$

Defining the **electric displacement**

$$\vec{D} := \epsilon_0 \vec{E} + \vec{P}, \quad \text{we have} \quad \nabla \cdot \vec{D} = \rho.$$

If the medium is isotropic,

$$\vec{P} = \epsilon_0 \chi_e \vec{E},$$

where χ_e is the **electric susceptibility**. Then

$$\vec{D} = \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E} = \epsilon_0 (1 + \chi_e) \vec{E} \equiv \epsilon \vec{E},$$

with the **electric permittivity** $\epsilon := \epsilon_0 (1 + \chi_e)$ and the **dielectric constant** $\epsilon/\epsilon_0 = 1 + \chi_e$. Thus $\nabla \cdot (\epsilon \vec{E}) = \rho$, and if the material is also uniform, $\nabla \cdot \vec{E} = \rho/\epsilon$. The boundary conditions are

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{n}_{21} = \sigma, \quad (\vec{E}_2 - \vec{E}_1) \times \hat{n}_{21} = 0.$$

4.7 Electrostatic Energy in Dielectric Media

A small change $\delta\rho$ of the free charge density changes the energy by

$$\delta W = \int d^3x \delta\rho \phi = \int d^3x \nabla \cdot (\delta \vec{D}) \phi = \int d^3x \vec{E} \cdot \delta \vec{D},$$

where we integrated by parts and used $\vec{E} = -\nabla\phi$. Bringing the displacement from 0 to its final value $\vec{D}$,

$$W = \int d^3x \int_0^{\vec{D}} \vec{E} \cdot \delta \vec{D}.$$

If the medium is linear, $\vec{E} \cdot \delta \vec{D} = \frac{1}{2} \delta(\vec{E} \cdot \vec{D})$, and

$$W = \frac{1}{2} \int d^3x \vec{E} \cdot \vec{D}.$$

5 Magnetostatics, Faraday's Law, Quasi-Static Fields

5.1 Introduction and Definitions

Conservation of charge demands

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0 :$$

if the charge in a region of space changes, there must be an influx or outflux of charge. Steady-state systems (magnetostatics) require

$$\nabla \cdot \vec{J} = 0.$$

5.2 Biot and Savart Law

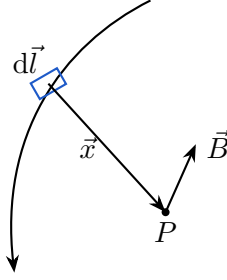


Figure 5.1: Contribution of a current element $I d\vec{l}$ to the field at P .

$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \vec{x}}{|\vec{x}|^3}, \quad I d\vec{l} \rightarrow \vec{J} d^3x.$$

The force on a current I due to a magnetic induction $\vec{B}$ is

$$d\vec{F} = I d\vec{l} \times \vec{B},$$

and in general

$$\vec{F} = \int d^3x \vec{J}(\vec{x}) \times \vec{B}(\vec{x}), \quad \vec{N} = \int d^3x \vec{x} \times (\vec{J} \times \vec{B}).$$

5.3 Differential Equations of Magnetostatics and Ampère's Law

$$\vec{B}(\vec{x}) = \frac{\mu_0}{4\pi} \int d^3x' \vec{J}(\vec{x}') \times \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} = \frac{\mu_0}{4\pi} \nabla \times \int d^3x' \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|}.$$

This means that

$$\nabla \cdot \vec{B} = 0 :$$

there are no magnetic monopoles. Taking the curl, and using $\nabla \times (\nabla \times \vec{a}) = \nabla(\nabla \cdot \vec{a}) - \Delta \vec{a}$,

$$\nabla \times \vec{B} = \frac{\mu_0}{4\pi} \nabla \int d^3x' \vec{J}(\vec{x}') \cdot \nabla \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) - \frac{\mu_0}{4\pi} \int d^3x' \vec{J}(\vec{x}') \Delta \left(\frac{1}{|\vec{x} - \vec{x}'|} \right).$$

With $\nabla(1/|\vec{x} - \vec{x}'|) = -\nabla'(1/|\vec{x} - \vec{x}'|)$ and $\Delta(1/|\vec{x} - \vec{x}'|) = -4\pi\delta(\vec{x} - \vec{x}')$,

$$\nabla \times \vec{B} = -\frac{\mu_0}{4\pi} \nabla \int d^3x' \vec{J}(\vec{x}') \cdot \nabla' \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) + \mu_0 \vec{J} = \mu_0 \vec{J} + \frac{\mu_0}{4\pi} \nabla \int d^3x' \frac{\nabla' \cdot \vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|},$$

where we integrated by parts. In the steady state $\nabla' \cdot \vec{J} = 0$, so

$$\nabla \times \vec{B} = \mu_0 \vec{J},$$

which in integral form is **Ampère's law**

$$\oint_{\partial S} d\vec{l} \cdot \vec{B} = \mu_0 I_{\text{enc}}.$$

5.4 Vector Potential

The basic differential laws of magnetostatics are

$$\nabla \times \vec{B} = \mu_0 \vec{J}, \quad \nabla \cdot \vec{B} = 0.$$

Notice that if $\vec{J} = 0$ we could make use of a **magnetic scalar potential** ϕ_M : since $\nabla \times \vec{B} = 0$, $\vec{B} = -\nabla \phi_M$ and $\Delta \phi_M = 0$. In general ($\vec{J} \neq 0$), since $\nabla \cdot \vec{B} = 0$, we can always write

$$\vec{B} = \nabla \times \vec{A}.$$

We have already computed $\vec{A}$ in Section 5.3:

$$\vec{B} = \nabla \times \left(\frac{\mu_0}{4\pi} \int d^3x' \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} \right) \implies \vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int d^3x' \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} + \nabla \psi.$$

The gradient term must be added because only $\nabla \times \vec{A}$ is defined, not its divergence, which we are free to choose. The magnetic induction is unaltered under

$$\vec{A} \rightarrow \vec{A} + \nabla \psi \quad (\nabla \times \nabla \psi = 0),$$

which is called a **gauge transformation**. Now

$$\nabla \times \vec{B} = \mu_0 \vec{J} \implies \nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \Delta \vec{A} = \mu_0 \vec{J}.$$

Since we have the freedom to choose the divergence of $\vec{A}$, we can choose $\nabla \cdot \vec{A} = 0$ (Coulomb gauge), and then

$$\Delta \vec{A} = -\mu_0 \vec{J}.$$

From the discussion in electrostatics, in unbounded space

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int d^3x' \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|}.$$

Since this $\vec{A}$ already satisfies $\nabla \cdot \vec{A} = 0$, we need $\Delta \psi = 0$; if this holds in all space (with $\nabla \psi$ vanishing at infinity), $\nabla \psi = 0$.

5.6 Magnetic Fields of a Localised Current Distribution, Magnetic Moment

Assuming $|\vec{x}| \gg |\vec{x}'|$,

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{1}{|\vec{x}|} + \frac{\vec{x} \cdot \vec{x}'}{|\vec{x}|^3} + \dots$$

Then

$$A_i(\vec{x}) = \frac{\mu_0}{4\pi} \left[\frac{1}{|\vec{x}|} \int d^3x' J_i(\vec{x}') + \frac{\vec{x}}{|\vec{x}|^3} \cdot \int d^3x' J_i(\vec{x}') \vec{x}' + \dots \right].$$

For a localised current and any well-behaved f, g ,

$$\int d^3x' (f \vec{J} \cdot \nabla' g + g \vec{J} \cdot \nabla' f + f g \nabla' \cdot \vec{J}) = 0.$$

- With $f = 1$, $g = x'_i$ and $\nabla \cdot \vec{J} = 0$: $\int d^3x' J_i(\vec{x}') = 0$. Therefore there is no monopole term in the expansion of A_i .
- With $f = x'_i$, $g = x'_j$ and $\nabla \cdot \vec{J} = 0$: $\int d^3x' (x'_i J_j + x'_j J_i) = 0$.

Therefore

$$\begin{aligned} \vec{x} \cdot \int d^3x' \vec{x}' J_i &= \sum_j x_j \int d^3x' x'_j J_i = -\frac{1}{2} \sum_j x_j \int d^3x' (x'_i J_j - x'_j J_i) \\ &= -\frac{1}{2} \sum_{j,k} \varepsilon_{ijk} x_j \int d^3x' (\vec{x}' \times \vec{J})_k = -\frac{1}{2} \left[\vec{x} \times \int d^3x' (\vec{x}' \times \vec{J}) \right]_i. \end{aligned}$$

Defining the **magnetic moment density** or **magnetisation**

$$\vec{\mathcal{M}} := \frac{1}{2} \vec{x} \times \vec{J}(\vec{x})$$

and the **magnetic moment**

$$\vec{m} := \frac{1}{2} \int d^3x' \vec{x}' \times \vec{J}(\vec{x}'),$$

the lowest non-vanishing term in the expansion is

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{x}}{|\vec{x}|^3} + \dots$$

The corresponding magnetic induction is

$$\vec{B}(\vec{x}) = \frac{\mu_0}{4\pi} \frac{3\hat{n}(\hat{n} \cdot \vec{m}) - \vec{m}}{|\vec{x}|^3}, \quad \hat{n} := \frac{\vec{x}}{|\vec{x}|},$$

exactly the same form as the electric field of a dipole.

Notice that if a current is confined to a plane loop,

$$\vec{m} = \frac{I}{2} \oint \vec{x} \times d\vec{l} \implies |\vec{m}| = I \cdot (\text{area}).$$

For point charges, writing $\vec{J} = \sum_i q_i \vec{v}_i \delta(\vec{x} - \vec{x}_i)$,

$$\vec{m} = \frac{1}{2} \sum_i q_i (\vec{x}_i \times \vec{v}_i) = \sum_i \frac{q_i}{2M_i} \vec{L}_i,$$

where M_i are the masses and $\vec{L}_i$ the orbital angular momenta. If all particles have the same ratio $q_i/M_i = e/M$,

$$\vec{m} = \frac{e}{2M} \vec{L}.$$

This is the well-known **classical connection between angular momentum and magnetic moment**, which holds for orbital motion even on the atomic scale.

5.7 Force and Torque on and Energy of a Localised Current Distribution in an External Magnetic Induction

If a localised distribution of current is placed in an external magnetic induction $\vec{B}(\vec{x})$, it experiences forces and torques according to Ampère's laws. If $\vec{B}$ varies slowly over the region of the current,

$$B_k(\vec{x}) = B_k(0) + \vec{x} \cdot \nabla B_k(0) + \dots,$$

and the force becomes

$$F_i = \sum_{j,k} \varepsilon_{ijk} \left[B_k(0) \underbrace{\int d^3x' J_j(\vec{x}')}_0 + \int d^3x' J_j(\vec{x}') \vec{x}' \cdot \nabla B_k(0) + \dots \right].$$

The lowest contribution comes from the second term:

$$F_i = \sum_{j,k} \varepsilon_{ijk} (\vec{m} \times \nabla)_j B_k, \quad \vec{F} = (\vec{m} \times \nabla) \times \vec{B} = \nabla(\vec{m} \cdot \vec{B}) - \underbrace{\vec{m} (\nabla \cdot \vec{B})}_0,$$

so that

$$\vec{F} = \nabla(\vec{m} \cdot \vec{B}).$$

This force represents the rate of change of the total mechanical momentum, including the “hidden mechanical momentum” associated with the presence of electromagnetic momentum.

The torque, using the leading term in the expansion, is

$$\vec{N} = \int d^3x' \vec{x}' \times (\vec{J} \times \vec{B}(0)) = \int d^3x' [(\vec{x}' \cdot \vec{B}(0)) \vec{J} - (\vec{x}' \cdot \vec{J}) \vec{B}(0)].$$

The second term vanishes (take $f = g = r'$ in the identity above), and the first gives

$$\vec{N} = \vec{m} \times \vec{B}(0).$$

The potential energy of a permanent magnetic moment (or dipole) in an external magnetic field is

$$U = -\vec{m} \cdot \vec{B} \quad (\text{since } \vec{F} = -\nabla U),$$

so the dipole tends to orient itself parallel to the field.

5.8 Macroscopic Equations, Boundary Conditions on $\vec{B}$ and $\vec{H}$

As with the electric field, we go from the microscopic to the macroscopic description. The equation $\nabla \cdot \vec{B} = 0$ does not change. On the other hand, the magnetisation is

$$\vec{M}(\vec{x}) = \sum_i N_i \langle \vec{m}_i \rangle,$$

and

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int d^3x' \left[\frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} + \frac{\vec{M}(\vec{x}') \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \right],$$

where the second term is $\vec{M}(\vec{x}') \times \nabla'(1/|\vec{x} - \vec{x}'|)$. Integrating by parts,

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int d^3x' \frac{\vec{J}(\vec{x}') + \nabla' \times \vec{M}(\vec{x}')}{|\vec{x} - \vec{x}'|},$$

where $\vec{J}_M = \nabla \times \vec{M}$ is an effective current density. Since $\nabla \times \vec{B}_{\text{micro}} = \mu_0 \vec{J}_{\text{micro}}$,

$$\nabla \times \vec{B} = \mu_0 [\vec{J} + \nabla \times \vec{M}].$$

Defining the **magnetic field**

$$\vec{H} := \frac{1}{\mu_0} \vec{B} - \vec{M},$$

the macroscopic equations are

$$\nabla \times \vec{H} = \vec{J}, \quad \nabla \cdot \vec{B} = 0 \quad \parallel \quad \nabla \cdot \vec{D} = \rho, \quad \nabla \times \vec{E} = 0.$$

For isotropic diamagnetic or paramagnetic materials,

$$\vec{B} = \mu \vec{H},$$

where μ is the **magnetic permeability**:

- $\mu > \mu_0$: paramagnetic (linear);
- $\mu < \mu_0$: diamagnetic (linear);
- $\vec{B} = \vec{F}(\vec{H})$: ferromagnetic (non-linear), which shows **hysteresis**.

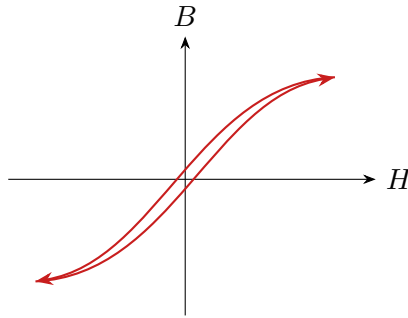


Figure 5.2: Hysteresis loop of a ferromagnet.

The boundary conditions are

$$(\vec{B}_2 - \vec{B}_1) \cdot \hat{n} = 0, \quad \hat{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}.$$

5.15 Faraday's Law of Induction

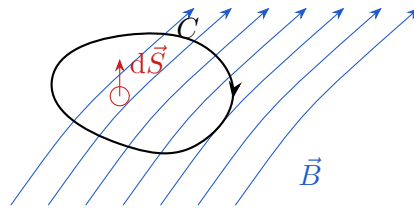


Figure 5.3: A circuit C bounding a surface S in a magnetic induction $\vec{B}$.

Faraday's law states

$$\oint_{\partial S} d\vec{l} \cdot \vec{E} = -\frac{d}{dt} \int_S d\vec{S} \cdot \vec{B},$$

where the proportionality constant (here 1 in SI units) is fixed by invariance under Galilean transformations. Using Stokes's theorem for a fixed circuit,

$$\int_S d\vec{S} \cdot \nabla \times \vec{E} = -\frac{d}{dt} \int_S d\vec{S} \cdot \vec{B} \implies \int_S d\vec{S} \cdot \left(\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} \right) = 0,$$

and since S is arbitrary,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}.$$

5.16 Energy in the Magnetic Field

For a circuit with current I , a change δF of the magnetic flux $F = \int_S d\vec{S} \cdot \vec{B}$ through it requires the work

$$\delta W = I \delta F.$$

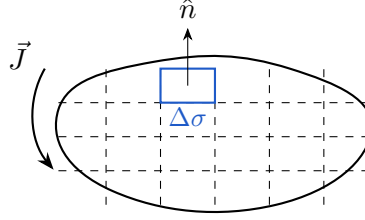


Figure 5.4: A current distribution decomposed into elementary current loops (tubes of cross section $\Delta\sigma$).

Decomposing the current distribution into elementary loops of cross section $\Delta\sigma$ carrying current $J\Delta\sigma$,

$$\Delta(\delta W) = (J\Delta\sigma) \int_S d\vec{S} \cdot \delta\vec{B} = J\Delta\sigma \int_S d\vec{S} \cdot \nabla \times (\delta\vec{A}) = J\Delta\sigma \oint d\vec{l} \cdot \delta\vec{A},$$

and summing over all loops,

$$\begin{aligned} \delta W &= \int d^3x \vec{J} \cdot \delta\vec{A} = \int d^3x (\nabla \times \vec{H}) \cdot \delta\vec{A} \\ &= \int d^3x [\vec{H} \cdot (\nabla \times \delta\vec{A}) + \nabla \cdot (\vec{H} \times \delta\vec{A})] = \int d^3x \vec{H} \cdot \delta\vec{B}, \end{aligned}$$

where the divergence term vanishes for a localised field distribution. For paramagnetic or diamagnetic (linear) materials, $\vec{H} \cdot \delta\vec{B} = \frac{1}{2}\delta(\vec{H} \cdot \vec{B})$, and

$$W = \frac{1}{2} \int d^3x \vec{H} \cdot \vec{B},$$

or, assuming a linear relation between $\vec{J}$ and $\vec{A}$,

$$W = \frac{1}{2} \int d^3x \vec{J} \cdot \vec{A}.$$

6 Maxwell Equations, Macroscopic Electromagnetism, Conservation Laws

6.1 Maxwell's Displacement Current; Maxwell Equations

$$\begin{aligned} \nabla \cdot \vec{D} &= \rho && \text{(Gauss),} \\ \nabla \times \vec{H} &= \vec{J} + \frac{\partial \vec{D}}{\partial t} && \text{(Ampère–Maxwell),} \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} && \text{(Faraday),} \\ \nabla \cdot \vec{B} &= 0 && \text{(absence of free magnetic monopoles).} \end{aligned}$$

The term $\partial\vec{D}/\partial t$ is the **displacement current**, introduced by Maxwell.

6.2 Vector and Scalar Potentials

It is often convenient to introduce potentials, obtaining a smaller number of second-order equations. The homogeneous equations act as constraints:

$$\nabla \cdot \vec{B} = 0 \implies \vec{B} = \nabla \times \vec{A}, \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \implies \nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t} \right) = 0,$$

so that $\vec{E} + \partial \vec{A} / \partial t = -\nabla \phi$, i.e.

$$\vec{E} = -\nabla \phi - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \nabla \times \vec{A}.$$

Replacing the fields in terms of the potentials in the other Maxwell equations (in vacuum),

$$\begin{aligned} \Delta \phi + \frac{\partial}{\partial t} \nabla \cdot \vec{A} &= -\frac{\rho}{\varepsilon_0}, \\ \Delta \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \nabla \left(\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right) &= -\mu_0 \vec{J}. \end{aligned}$$

These are **two equations instead of four**; however, they are coupled. We make use of the arbitrariness in the definition of $\vec{A}$: since $\vec{B} = \nabla \times \vec{A}$, the change $\vec{A} \rightarrow \vec{A} + \nabla \Lambda$ leaves $\vec{B}$ unchanged, but not $\vec{E}$; to fix that we also need $\phi \rightarrow \phi - \partial \Lambda / \partial t$. This freedom,

$$\vec{A} \rightarrow \vec{A} + \nabla \Lambda, \quad \phi \rightarrow \phi - \frac{\partial \Lambda}{\partial t},$$

is called **gauge freedom** (or gauge transformation), and can be used to impose the **Lorenz condition**

$$\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0,$$

which uncouples the equations for the potentials into **wave equations with source terms**:

$$\Delta \phi - \frac{1}{c^2} \partial_t^2 \phi = -\frac{\rho}{\varepsilon_0}, \quad \Delta \vec{A} - \frac{1}{c^2} \partial_t^2 \vec{A} = -\mu_0 \vec{J}.$$

6.3 Gauge Transformations, Lorenz Gauge, Coulomb Gauge

The transformation

$$\vec{A} \rightarrow \vec{A}' = \vec{A} + \nabla \Lambda, \quad \phi \rightarrow \phi' = \phi - \partial_t \Lambda$$

is called a **gauge transformation**, and the invariance of the fields under such transformations is called **gauge invariance**. To satisfy the Lorenz condition,

$$0 = \nabla \cdot \vec{A}' + \frac{1}{c^2} \partial_t \phi' = \nabla \cdot \vec{A} + \frac{1}{c^2} \partial_t \phi + \Delta \Lambda - \frac{1}{c^2} \partial_t^2 \Lambda \implies \Delta \Lambda - \frac{1}{c^2} \partial_t^2 \Lambda = -\left(\nabla \cdot \vec{A} + \frac{1}{c^2} \partial_t \phi \right).$$

Provided we find a function Λ that satisfies this equation, the new potentials satisfy the Lorenz condition. Even for potentials that satisfy the Lorenz condition there is some arbitrariness: the **restricted gauge transformation** $\vec{A} \rightarrow \vec{A} + \nabla \Lambda$, $\phi \rightarrow \phi - \partial_t \Lambda$ with

$$\Delta \Lambda - \frac{1}{c^2} \partial_t^2 \Lambda = 0$$

preserves it. All potentials in this restricted class are said to belong to the **Lorenz gauge**. In it

- ϕ and $\vec{A}$ are on equivalent footings;

- they satisfy uncoupled wave equations;
- the condition is independent of the coordinate system chosen (it fits naturally into special relativity).

Another useful gauge is the **Coulomb, radiation** or **transverse gauge**:

$$\nabla \cdot \vec{A} = 0.$$

Then

$$\Delta\phi = -\frac{\rho}{\varepsilon_0} \implies \phi(\vec{x}, t) = \frac{1}{4\pi\varepsilon_0} \int d^3x' \frac{\rho(\vec{x}', t)}{|\vec{x} - \vec{x}'|},$$

which is just the instantaneous Coulomb potential. On the other hand, the vector potential satisfies

$$\Delta\vec{A} - \frac{1}{c^2} \partial_t^2 \vec{A} = -\mu_0 \vec{J} + \frac{1}{c^2} \nabla \partial_t \phi,$$

where the last term is irrotational (longitudinal), since $\nabla \times (\nabla \partial_t \phi) = 0$. Decompose the current into longitudinal and transverse parts,

$$\vec{J} = \vec{J}_l + \vec{J}_t, \quad \nabla \times \vec{J}_l = 0, \quad \nabla \cdot \vec{J}_t = 0.$$

Starting from $\nabla \times (\nabla \times \vec{J}) = \nabla(\nabla \cdot \vec{J}) - \Delta\vec{J}$ together with $\Delta(1/|\vec{x} - \vec{x}'|) = -4\pi\delta(\vec{x} - \vec{x}')$, one finds

$$\vec{J}_l = -\frac{1}{4\pi} \nabla \int d^3x' \frac{\nabla' \cdot \vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|}, \quad \vec{J}_t = \frac{1}{4\pi} \nabla \times \nabla \times \int d^3x' \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|}.$$

Using the continuity equation, $\nabla \cdot \vec{J} = -\partial_t \rho$, together with $\nabla \cdot (\nabla \phi) = -\rho/\varepsilon_0$,

$$\nabla \cdot \vec{J}_l = \varepsilon_0 \partial_t \nabla \cdot (\nabla \phi) \implies \mu_0 \vec{J}_l = \frac{1}{c^2} \nabla \partial_t \phi,$$

and therefore

$$\Delta\vec{A} - \frac{1}{c^2} \partial_t^2 \vec{A} = -\mu_0 \vec{J}_t, \quad \phi(\vec{x}, t) = \frac{1}{4\pi\varepsilon_0} \int d^3x' \frac{\rho(\vec{x}', t)}{|\vec{x} - \vec{x}'|}.$$

Notice that when no sources are present, $\phi = 0$ in the Coulomb gauge and

$$\vec{E} = -\partial_t \vec{A}, \quad \vec{B} = \nabla \times \vec{A}.$$

6.4 Green Functions for the Wave Equation

The basic structure of the wave equation is

$$\Delta\Psi - \frac{1}{c^2} \partial_t^2 \Psi = -4\pi f(\vec{x}, t),$$

where c is the velocity of propagation in the medium and f a known source distribution. Let us consider a simple situation with no boundaries and apply a Fourier transformation in time:

$$\Psi(\vec{x}, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega \Psi(\vec{x}, \omega) e^{-i\omega t}, \quad f(\vec{x}, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega f(\vec{x}, \omega) e^{-i\omega t},$$

with inverse transformations

$$\Psi(\vec{x}, \omega) = \int_{-\infty}^{\infty} dt \Psi(\vec{x}, t) e^{i\omega t}, \quad f(\vec{x}, \omega) = \int_{-\infty}^{\infty} dt f(\vec{x}, t) e^{i\omega t}.$$

The wave equation is transformed into the **inhomogeneous Helmholtz wave equation**

$$(\Delta + k^2) \Psi(\vec{x}, \omega) = -4\pi f(\vec{x}, \omega),$$

where $k := \omega/c$ is the wave number associated with the frequency ω (causality will impose some restrictions). The associated Green function satisfies

$$(\Delta + k^2) G_k(\vec{x}, \vec{x}') = -4\pi \delta(\vec{x} - \vec{x}').$$

If there are no boundaries, the Green function can depend only on $\vec{R} = \vec{x} - \vec{x}'$, and must in fact be spherically symmetric, i.e. depend only on $R = |\vec{R}|$:

$$\frac{1}{R} \frac{d^2}{dR^2} (RG_k) + k^2 G_k = -4\pi \delta(\vec{R}).$$

Everywhere except at $R = 0$,

$$RG_k = A e^{ikR} + B e^{-ikR}.$$

In the limit $kR \rightarrow 0$ the equation becomes the Poisson equation, so $\lim_{kR \rightarrow 0} G_k(R) = 1/R$. Hence

$$G_k(R) = A G_k^{(+)}(R) + B G_k^{(-)}(R), \quad G_k^{(\pm)}(R) = \frac{e^{\pm ikR}}{R}, \quad A + B = 1,$$

where $G_k^{(+)}$ ($G_k^{(-)}$) represents a diverging (converging) spherical wave. A and B depend on the boundary conditions in time.

The time-dependent Green function satisfies

$$\left(\Delta - \frac{1}{c^2} \partial_t^2 \right) G^{(\pm)}(\vec{x}, t; \vec{x}', t') = -4\pi \delta(\vec{x} - \vec{x}') \delta(t - t').$$

Using $\delta(t - t') = \frac{1}{2\pi} \int d\omega e^{-i\omega(t-t')}$,

$$G^{(\pm)}(R, \tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega \frac{e^{\pm ikR}}{R} e^{-i\omega\tau},$$

where $\tau = t - t'$ is the relative time. For a non-dispersive medium, $k = \omega/c$ and

$$G^{(\pm)}(R, \tau) = \frac{1}{R} \delta\left(\tau \mp \frac{R}{c}\right),$$

where R/c is the time of propagation of the disturbance. $G^{(+)}$ is the **retarded Green function** (the effect observed at time t was produced a distance R away at the earlier time $t' = t - R/c$) and $G^{(-)}$ the **advanced Green function**. More explicitly,

$$G^{(\pm)}(\vec{x}, t; \vec{x}', t') = \frac{\delta\left(t' - \left(t \mp \frac{|\vec{x} - \vec{x}'|}{c}\right)\right)}{|\vec{x} - \vec{x}'|}.$$

Solutions of the inhomogeneous wave equation are

$$\Psi^{(\pm)}(\vec{x}, t) = \int d^3x' dt' G^{(\pm)}(\vec{x}, t; \vec{x}', t') f(\vec{x}', t').$$

If at remote past times, $t \rightarrow -\infty$, there is a wave $\Psi_{\text{in}}(\vec{x}, t)$ satisfying the homogeneous wave equation, and around $t' = 0$ a source distribution $f(\vec{x}', t') \neq 0$ acts,

$$\Psi(\vec{x}, t) = \Psi_{\text{in}}(\vec{x}, t) + \int d^3x' dt' G^{(+)}(\vec{x}, t; \vec{x}', t') f(\vec{x}', t') = \Psi_{\text{in}}(\vec{x}, t) + \int d^3x' \frac{[f(\vec{x}', t')]_{\text{ret}}}{|\vec{x} - \vec{x}'|}.$$

The presence of $G^{(+)}$ guarantees that at remote times t , before the source has been activated, there is no contribution from the integral. Similarly, if there is a wave $\Psi_{\text{out}}(\vec{x}, t)$ at remote late times $t \rightarrow +\infty$ that satisfies the homogeneous wave equation,

$$\Psi(\vec{x}, t) = \Psi_{\text{out}}(\vec{x}, t) + \int d^3x' dt' G^{(-)}(\vec{x}, t; \vec{x}', t') f(\vec{x}', t'),$$

and no signal from the source exists explicitly after the source shuts off.

6.5 Retarded Solutions for the Fields: Jefimenko's Generalisations of the Coulomb and Biot–Savart Laws; Heaviside–Feynman Expressions for a Point Charge

In the Lorenz gauge,

$$\phi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{[\rho(\vec{x}', t')]_{\text{ret}}}{|\vec{x} - \vec{x}'|}, \quad \vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int d^3x' \frac{[\vec{J}(\vec{x}', t')]_{\text{ret}}}{|\vec{x} - \vec{x}'|},$$

with $\vec{R} = \vec{x} - \vec{x}'$ and $\hat{R} := \vec{R}/R$. From the Maxwell equations the fields themselves satisfy

$$\Delta \vec{E} - \frac{1}{c^2} \partial_t^2 \vec{E} = -\frac{1}{\epsilon_0} \left(-\nabla \rho - \frac{1}{c^2} \partial_t \vec{J} \right), \quad \Delta \vec{B} - \frac{1}{c^2} \partial_t^2 \vec{B} = -\mu_0 \nabla \times \vec{J},$$

so that

$$\vec{E}(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{R} \left[-\nabla' \rho - \frac{1}{c^2} \partial_{t'} \vec{J} \right]_{\text{ret}}, \quad \vec{B}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int d^3x' \frac{1}{R} [\nabla' \times \vec{J}]_{\text{ret}}.$$

Since $[f(\vec{x}', t')]_{\text{ret}} = f(\vec{x}', t - |\vec{x} - \vec{x}'|/c)$, the retardation does not commute with ∇' : $[\nabla' f]_{\text{ret}} \neq \nabla' [f]_{\text{ret}}$. With $t' = t - R/c$ and $\nabla' t' = \hat{R}/c$,

$$\begin{aligned} [\nabla' \rho]_{\text{ret}} &= \nabla' [\rho]_{\text{ret}} - \frac{\hat{R}}{c} [\partial_{t'} \rho]_{\text{ret}}, \\ [\nabla' \times \vec{J}]_{\text{ret}} &= \nabla' \times [\vec{J}]_{\text{ret}} + \frac{1}{c} [\partial_{t'} \vec{J}]_{\text{ret}} \times \hat{R}. \end{aligned}$$

Integrating by parts, we obtain **Jefimenko's generalisations**:

$$\begin{aligned} \vec{E}(\vec{x}, t) &= \frac{1}{4\pi\epsilon_0} \int d^3x' \left\{ \frac{\hat{R}}{R^2} [\rho(\vec{x}', t')]_{\text{ret}} + \frac{\hat{R}}{cR} [\partial_{t'} \rho(\vec{x}', t')]_{\text{ret}} - \frac{1}{c^2 R} [\partial_{t'} \vec{J}(\vec{x}', t')]_{\text{ret}} \right\}, \\ \vec{B}(\vec{x}, t) &= \frac{\mu_0}{4\pi} \int d^3x' \left\{ [\vec{J}(\vec{x}', t')]_{\text{ret}} \times \frac{\hat{R}}{R^2} + [\partial_{t'} \vec{J}(\vec{x}', t')]_{\text{ret}} \times \frac{\hat{R}}{cR} \right\}. \end{aligned}$$

Considering that $t' = t - |\vec{x} - \vec{x}'|/c$, we have $[\partial_{t'} f(\vec{x}', t')]_{\text{ret}} = \partial_t [f(\vec{x}', t')]_{\text{ret}}$. Thus, for a point charge, with $\rho(\vec{x}', t') = q \delta(\vec{x}' - \vec{r}_0(t'))$ and $\vec{J}(\vec{x}', t') = \rho \vec{v}(t')$, we obtain the **Heaviside–Feynman expressions**

$$\begin{aligned} \vec{E} &= \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{R}}{\kappa R^2} \right]_{\text{ret}} + \frac{1}{c} \partial_t \left[\frac{\hat{R}}{\kappa R} \right]_{\text{ret}} - \frac{1}{c^2} \partial_t \left[\frac{\vec{v}}{\kappa R} \right]_{\text{ret}} \right\}, \\ \vec{B} &= \frac{\mu_0 q}{4\pi} \left\{ \left[\frac{\vec{v} \times \hat{R}}{\kappa R^2} \right]_{\text{ret}} + \frac{1}{c} \partial_t \left[\frac{\vec{v} \times \hat{R}}{\kappa R} \right]_{\text{ret}} \right\}, \end{aligned}$$

where

$$\kappa := 1 - \frac{\vec{v} \cdot \hat{R}}{c}$$

is the **retardation factor**.

6.7 Poynting's Theorem and Conservation of Energy and Momentum for a System of Charged Particles and Electromagnetic Fields

For a continuous distribution of charge and current, the total rate of doing work by the fields in a finite volume V is

$$\int_V d^3x \vec{J} \cdot \vec{E}.$$

In terms of the fields only, using the Maxwell equations,

$$\int_V d^3x \vec{J} \cdot \vec{E} = - \int_V d^3x \left[\nabla \cdot (\vec{E} \times \vec{H}) + \vec{E} \cdot \partial_t \vec{D} + \vec{H} \cdot \partial_t \vec{B} \right].$$

Assuming that

- the macroscopic medium is linear in its electric and magnetic properties,
- the total electromagnetic energy density is $u = \frac{1}{2}(\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H})$,

we get

$$- \int_V d^3x \vec{J} \cdot \vec{E} = \int_V d^3x \left[\partial_t u + \nabla \cdot (\vec{E} \times \vec{H}) \right],$$

i.e. the **energy conservation law**

$$\partial_t u + \nabla \cdot \vec{S} = -\vec{J} \cdot \vec{E},$$

where

$$\vec{S} := \vec{E} \times \vec{H}$$

is the **Poynting vector**, representing the energy flow. If $dE_{\text{mech}}/dt = \int_V d^3x \vec{J} \cdot \vec{E}$, then

$$\frac{dE}{dt} = \frac{d}{dt}(E_{\text{mech}} + E_{\text{field}}) = - \oint_{\partial V} da \hat{n} \cdot \vec{S}, \quad E_{\text{field}} = \int_V d^3x u = \frac{\varepsilon_0}{2} \int_V d^3x (E^2 + c^2 B^2)$$

(the last expression in vacuum).

The conservation of linear momentum can be treated similarly. From the Lorentz force $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$,

$$\frac{d\vec{P}_{\text{mech}}}{dt} = \int_V d^3x (\rho \vec{E} + \vec{J} \times \vec{B}),$$

and, eliminating ρ and $\vec{J}$ with the Maxwell equations (in vacuum),

$$\frac{d\vec{P}_{\text{mech}}}{dt} + \frac{d}{dt} \int_V d^3x \varepsilon_0 (\vec{E} \times \vec{B}) = \varepsilon_0 \int_V d^3x \left[\vec{E}(\nabla \cdot \vec{E}) - \vec{E} \times (\nabla \times \vec{E}) + c^2 \vec{B}(\nabla \cdot \vec{B}) - c^2 \vec{B} \times (\nabla \times \vec{B}) \right].$$

The **electromagnetic momentum density** is

$$\vec{g} := \frac{1}{c^2} \vec{E} \times \vec{H} = \frac{1}{c^2} \vec{S},$$

and the right-hand side can be written as a divergence,

$$\frac{d}{dt} (\vec{P}_{\text{mech}} + \vec{P}_{\text{field}})_\alpha = \sum_\beta \int_V d^3x \frac{\partial}{\partial x_\beta} T_{\alpha\beta},$$

where

$$T_{\alpha\beta} := \varepsilon_0 \left[E_\alpha E_\beta + c^2 B_\alpha B_\beta - \frac{1}{2} (E^2 + c^2 B^2) \delta_{\alpha\beta} \right]$$

is the **Maxwell stress tensor**. By the divergence theorem,

$$\frac{d}{dt} (\vec{P}_{\text{mech}} + \vec{P}_{\text{field}})_\alpha = \oint_{\partial V} \sum_\beta T_{\alpha\beta} n_\beta da,$$

which expresses the **conservation of momentum**.

7 Plane Electromagnetic Waves and Wave Propagation

7.1 Plane Waves in a Non-Conducting Medium

In the absence of sources, and in a non-conducting medium,

$$\nabla \cdot \vec{B} = 0, \quad \nabla \times \vec{E} + \partial_t \vec{B} = 0, \quad \nabla \cdot \vec{D} = 0, \quad \nabla \times \vec{H} - \partial_t \vec{D} = 0.$$

Assuming solutions with harmonic time dependence $e^{-i\omega t}$, $\vec{F}(\vec{x}, t) = \vec{F}(\vec{x}, \omega) e^{-i\omega t}$ for $\vec{B}$, $\vec{E}$, $\vec{D}$ and $\vec{H}$,

$$\nabla \cdot \vec{B} = 0, \quad \nabla \times \vec{E} - i\omega \vec{B} = 0, \quad \nabla \cdot \vec{D} = 0, \quad \nabla \times \vec{H} + i\omega \vec{D} = 0.$$

For uniform and isotropic linear media, $\vec{D} = \varepsilon \vec{E}$ and $\vec{B} = \mu \vec{H}$, where ε and μ might be complex. For the present, $\varepsilon, \mu \in \mathbb{R}$ (no losses). Then

$$(\Delta + \mu\varepsilon\omega^2) \begin{Bmatrix} \vec{E} \\ \vec{B} \end{Bmatrix} = 0.$$

Considering as a possible solution a plane wave travelling in the x direction, $e^{ikx - i\omega t}$,

$$k = \sqrt{\mu\varepsilon} \omega.$$

The **phase velocity** of the wave is

$$v := \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n}, \quad n := \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}},$$

where n is the **index of refraction**. Using $\omega = kv$,

$$u_k(x, t) = a e^{ik(x-vt)} + b e^{-ik(x+vt)}.$$

If the medium is non-dispersive (ε and μ independent of the frequency), the general solution is

$$u(x, t) = f(x - vt) + g(x + vt);$$

if the medium is dispersive this is no longer a solution.

Let us consider an electromagnetic plane wave (which must satisfy the Maxwell equations):

$$\vec{E} = \vec{\mathcal{E}} e^{ik\hat{n}\cdot\vec{x} - i\omega t}, \quad \vec{B} = \vec{\mathcal{B}} e^{ik\hat{n}\cdot\vec{x} - i\omega t}.$$

From $\nabla \times \vec{E} - i\omega \vec{B} = 0$ and $\nabla \times \vec{B} + i\omega \mu \varepsilon \vec{E} = 0$ we get $(\Delta + \mu\varepsilon\omega^2)\{\vec{E}, \vec{B}\} = 0$, so

$$k^2 \hat{n} \cdot \hat{n} = \mu\varepsilon\omega^2, \quad \hat{n} \cdot \hat{n} = 1 \implies k = \omega \sqrt{\mu\varepsilon}.$$

Since $\nabla \cdot \vec{E} = 0 = \nabla \cdot \vec{B}$,

$$\hat{n} \cdot \vec{\mathcal{E}} = 0, \quad \hat{n} \cdot \vec{\mathcal{B}} = 0 :$$

the electric and magnetic fields are **perpendicular to the direction of propagation**, i.e. the wave is **transverse**. Moreover,

$$\vec{B} = -\frac{i}{\omega} \nabla \times \vec{E} \implies \vec{B} = \frac{k}{\omega} \hat{n} \times \vec{\mathcal{E}} = \sqrt{\mu\varepsilon} \hat{n} \times \vec{\mathcal{E}} = \frac{n}{c} \hat{n} \times \vec{\mathcal{E}},$$

so the electric and magnetic fields are orthogonal to each other. With orthonormal polarisation vectors $\hat{e}_1, \hat{e}_2$ perpendicular to $\hat{n}$,

$$\vec{\mathcal{E}} = E_0 \hat{e}_1, \quad \vec{\mathcal{B}} = \frac{n}{c} E_0 \hat{e}_2.$$

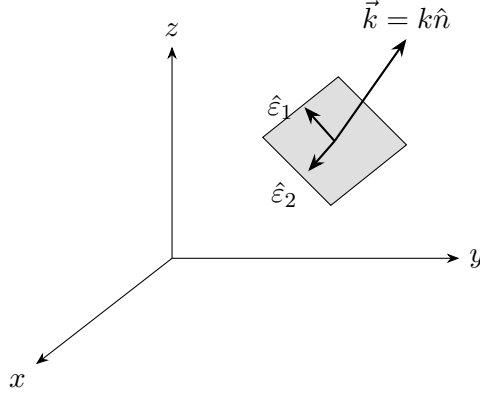


Figure 7.1: Wave vector and polarisation vectors of a plane wave.

The time-averaged Poynting vector is the real part of

$$\vec{S} = \frac{1}{2} \vec{E} \times \vec{H}^*, \quad \vec{S} = \frac{1}{2} \sqrt{\frac{\epsilon}{\mu}} |E_0|^2 \hat{n},$$

and the time-averaged energy density is

$$u = \frac{1}{4} \left(\epsilon \vec{E} \cdot \vec{E}^* + \frac{1}{\mu} \vec{B} \cdot \vec{B}^* \right) = \frac{\epsilon}{2} |E_0|^2.$$

7.2 Linear and Circular Polarisation; Stokes Parameters

A plane wave whose electric field vector is always in the direction of, say, $\hat{\epsilon}_1$ is said to be **linearly polarised**. Consider two linearly polarised waves,

$$\vec{E}_1 = E_1 e^{i\vec{k} \cdot \vec{x} - i\omega t} \hat{\epsilon}_1, \quad \vec{E}_2 = E_2 e^{i\vec{k} \cdot \vec{x} - i\omega t} \hat{\epsilon}_2, \quad \hat{\epsilon}_1 \cdot \hat{\epsilon}_2 = 0,$$

$$\vec{B}_j = \sqrt{\mu\epsilon} \frac{\vec{k} \times \vec{E}_j}{k}, \quad j = 1, 2.$$

The most general plane wave propagating in the direction of $\vec{k}$ is then

$$\vec{E} = (E_1 \hat{\epsilon}_1 + E_2 \hat{\epsilon}_2) e^{i\vec{k} \cdot \vec{x} - i\omega t}, \quad E_1, E_2 \in \mathbb{C}.$$

- If E_1 and E_2 have the *same phase*, $\vec{E}$ is a **linearly polarised** wave.
- If E_1 and E_2 have *different phases*, $\vec{E}$ is **elliptically polarised**.

If E_1 and E_2 have the same magnitude but a phase difference of $\pi/2$,

$$\vec{E} = E_0 (\hat{\epsilon}_1 \pm i\hat{\epsilon}_2) e^{i\vec{k} \cdot \vec{x} - i\omega t}.$$

Choosing $\hat{\epsilon}_1 = \hat{x}$, $\hat{\epsilon}_2 = \hat{y}$ and $\vec{k} = k\hat{z}$, the real fields are

$$E_x = E_0 \cos(kz - \omega t), \quad E_y = E_0 \cos\left(kz - \omega t \mp \frac{\pi}{2}\right) = \mp E_0 \sin(kz - \omega t).$$

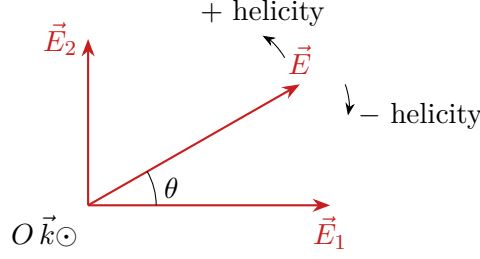


Figure 7.2: Circular polarisation, looking into the oncoming wave ($\vec{k}$ out of the page).

If $\vec{E}$ rotates counterclockwise (seen by an observer facing the oncoming wave), the wave has **positive helicity**; if it rotates clockwise, it has **negative helicity**.

Two circularly polarised waves form an equally acceptable set of basis fields for the description of a general state of polarisation:

$$\vec{\epsilon}_{\pm} = \frac{1}{\sqrt{2}}(\hat{\epsilon}_1 \pm i\hat{\epsilon}_2),$$

with the properties

$$\vec{\epsilon}_{\pm}^* \cdot \vec{\epsilon}_{\mp} = 0, \quad \vec{\epsilon}_{\pm}^* \cdot \hat{\epsilon}_3 = 0, \quad \vec{\epsilon}_{\pm}^* \cdot \vec{\epsilon}_{\pm} = 1,$$

where $\hat{\epsilon}_3 = \hat{n}$. Then $\vec{E} = (E_+\vec{\epsilon}_+ + E_-\vec{\epsilon}_-)e^{i\vec{k}\cdot\vec{x}-i\omega t}$. If $|E_+| \neq |E_-|$ but they have the same phase, $\vec{E}$ represents an elliptically polarised wave with principal axes along $\hat{\epsilon}_1$ and $\hat{\epsilon}_2$.

7.3 Reflection and Refraction of Electromagnetic Waves at a Plane Interface Between Dielectrics

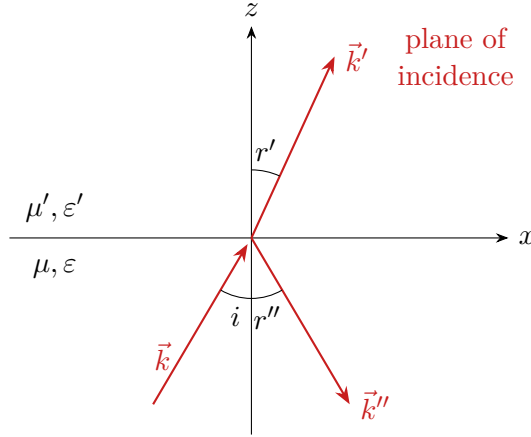


Figure 7.3: Incident ($\vec{k}$), refracted ($\vec{k}'$) and reflected ($\vec{k}''$) waves at the interface $z = 0$.

$$\begin{aligned} \text{Incident: } \vec{E} &= \vec{E}_0 e^{i\vec{k}\cdot\vec{x}-i\omega t}, & \vec{B} &= \sqrt{\mu\epsilon} \frac{\vec{k} \times \vec{E}}{k}, \\ \text{Refracted: } \vec{E}' &= \vec{E}'_0 e^{i\vec{k}'\cdot\vec{x}-i\omega t}, & \vec{B}' &= \sqrt{\mu'\epsilon'} \frac{\vec{k}' \times \vec{E}'}{k'}, \\ \text{Reflected: } \vec{E}'' &= \vec{E}''_0 e^{i\vec{k}''\cdot\vec{x}-i\omega t}, & \vec{B}'' &= \sqrt{\mu\epsilon} \frac{\vec{k}'' \times \vec{E}''}{k}, \end{aligned}$$

with

$$|\vec{k}| = |\vec{k}''| = \omega\sqrt{\mu\epsilon}, \quad |\vec{k}'| = \omega\sqrt{\mu'\epsilon'}.$$

All phase factors must be equal at $z = 0$:

$$\vec{k} \cdot \vec{x}|_{z=0} = \vec{k}' \cdot \vec{x}|_{z=0} = \vec{k}'' \cdot \vec{x}|_{z=0} \implies k \sin i = k' \sin r' = k'' \sin r''.$$

Since $k = k''$, $i = r''$: the angle of incidence equals the angle of reflection. Moreover,

$$\frac{k}{k'} = \frac{\sin r'}{\sin i} = \sqrt{\frac{\mu \varepsilon}{\mu' \varepsilon'}} = \frac{n}{n'}, \quad n = c\sqrt{\mu \varepsilon},$$

which is **Snell's law**

$$n \sin i = n' \sin r'.$$

Using now the boundary conditions with $\sigma = 0$ and $\vec{K} = 0$,

$$\begin{aligned} [\varepsilon(\vec{E}_0 + \vec{E}_0'') - \varepsilon' \vec{E}_0'] \cdot \hat{n} &= 0, \\ (\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}_0'' - \vec{k}' \times \vec{E}_0') \cdot \hat{n} &= 0, \\ (\vec{E}_0 + \vec{E}_0'' - \vec{E}_0') \times \hat{n} &= 0, \\ \left[\frac{1}{\mu}(\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}_0'') - \frac{1}{\mu'}(\vec{k}' \times \vec{E}_0') \right] \times \hat{n} &= 0. \end{aligned}$$

These give the **Fresnel equations**.

$\vec{E}$ perpendicular to the plane of incidence:

$$\frac{E_0'}{E_0} = \frac{2n \cos i}{n \cos i + \frac{\mu}{\mu'} \sqrt{n'^2 - n^2 \sin^2 i}}, \quad \frac{E_0''}{E_0} = \frac{n \cos i - \frac{\mu}{\mu'} \sqrt{n'^2 - n^2 \sin^2 i}}{n \cos i + \frac{\mu}{\mu'} \sqrt{n'^2 - n^2 \sin^2 i}}.$$

$\vec{E}$ parallel to the plane of incidence:

$$\frac{E_0'}{E_0} = \frac{2nn' \cos i}{\frac{\mu}{\mu'} n'^2 \cos i + n \sqrt{n'^2 - n^2 \sin^2 i}}, \quad \frac{E_0''}{E_0} = \frac{\frac{\mu}{\mu'} n'^2 \cos i - n \sqrt{n'^2 - n^2 \sin^2 i}}{\frac{\mu}{\mu'} n'^2 \cos i + n \sqrt{n'^2 - n^2 \sin^2 i}}.$$

7.8 Superposition of Waves in One Dimension; Group Velocity

In general electromagnetic waves are not monochromatic, and $\omega = \omega(k) = \omega(-k)$. Since the Maxwell equations are linear, the general solution (in one dimension) is of the form

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk A(k) e^{ikx - i\omega(k)t}, \quad A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx u(x, 0) e^{-ikx}.$$

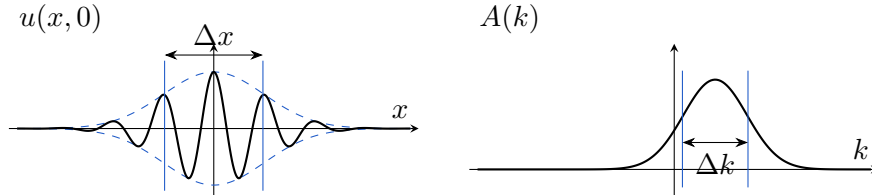


Figure 7.4: A pulse $u(x, 0)$ and its wave-number spectrum $A(k)$, with rms widths Δx and Δk .

Defining the rms deviations with weights $|u(x, 0)|^2$ and $|A(k)|^2$,

$$(\Delta x)^2 = \frac{\int dx (x - \bar{x})^2 |u(x, 0)|^2}{\int dx |u(x, 0)|^2}, \quad (\Delta k)^2 = \frac{\int dk (k - \bar{k})^2 |A(k)|^2}{\int dk |A(k)|^2},$$

one finds

$$\Delta x \Delta k \geq \frac{1}{2},$$

with equality for a Gaussian pulse.

The propagation of a pulse which is not too broad in its wave-number spectrum, or of a pulse in a medium for which the frequency depends weakly on the wave number, can be handled as follows. If $A(k)$ is fairly sharply peaked around k_0 ,

$$\omega(k) \simeq \omega_0 + \left. \frac{d\omega}{dk} \right|_{k_0} (k - k_0),$$

and

$$\begin{aligned} u(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk A(k) e^{ikx - i\omega t} \simeq \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk A(k) e^{ikx - i\left(\omega_0 + \left. \frac{d\omega}{dk} \right|_{k_0} (k - k_0)\right)t} \\ &= \frac{e^{i\left(k_0 \left. \frac{d\omega}{dk} \right|_{k_0} - \omega_0\right)t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk A(k) e^{i\left(x - t \left. \frac{d\omega}{dk} \right|_{k_0}\right)k}, \end{aligned}$$

that is,

$$u(x, t) \simeq u\left(x - t \left. \frac{d\omega}{dk} \right|_{k_0}, 0\right) e^{i\left(k_0 \left. \frac{d\omega}{dk} \right|_{k_0} - \omega_0\right)t} :$$

apart from a phase factor, the pulse travels along undistorted in shape. Its velocity is called the **group velocity**,

$$v_g = \left. \frac{d\omega}{dk} \right|_{k_0},$$

and in general $v_p = \omega/k \neq v_g$. For light waves, $\omega(k) = ck/n(k)$, so $v_p = c/n(k)$ and

$$\frac{d\omega}{dk} = \frac{c}{n} - \frac{ck}{n^2} \frac{dn}{dk} = \frac{c}{n} - \frac{\omega}{n} \frac{dn}{d\omega} v_g \implies v_g = \frac{c}{n(\omega) + \omega \frac{dn}{d\omega}}.$$

8 Waveguides, Resonant Cavities, and Optical Fibres

8.1 Fields at the Surface of and Within a Conductor

The boundary conditions at the surface of a conductor are

$$\hat{n} \cdot \vec{D} = \Sigma \text{ (surface charge density)}, \quad \hat{n} \times \vec{H} = \vec{K}, \quad \hat{n} \cdot (\vec{B} - \vec{B}_c) = 0, \quad \hat{n} \times (\vec{E} - \vec{E}_c) = 0.$$

However, for a finite conductivity σ , $\vec{J} = \sigma \vec{E}_c$ and there is no true surface current ($\vec{K} = 0$), so

$$\hat{n} \times (\vec{H} - \vec{H}_c) = 0.$$

Inside the conductor (neglecting the displacement current), $\nabla \times \vec{H}_c = \vec{J} = \sigma \vec{E}_c$ and $\nabla \times \vec{E}_c = -\partial_t \vec{B}_c = i\mu_c \omega \vec{H}_c$, so

$$\vec{E}_c = \frac{1}{\sigma} \nabla \times \vec{H}_c, \quad \vec{H}_c = -\frac{i}{\mu_c \omega} \nabla \times \vec{E}_c.$$

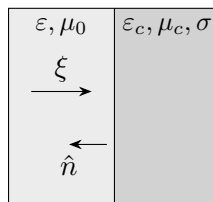


Figure 8.1: Surface of a good conductor. $\hat{n}$ points out of the conductor and ξ is the depth into it.

Near the surface the fields vary rapidly along the normal, so $\nabla \simeq -\hat{n} \partial_\xi$, where ξ is the normal coordinate into the conductor. Then

$$\vec{E}_c \simeq -\frac{1}{\sigma} \hat{n} \times \partial_\xi \vec{H}_c, \quad \vec{H}_c \simeq \frac{i}{\mu_c \omega} \hat{n} \times \partial_\xi \vec{E}_c \implies \hat{n} \cdot \vec{H}_c \simeq 0,$$

so $\vec{H}_c = \vec{H}_\parallel$ is essentially tangential. Moreover,

$$\hat{n} \times \vec{H}_c \simeq \frac{i}{\mu_c \omega} \hat{n} \times (\hat{n} \times \partial_\xi \vec{E}_c) = -\frac{i}{\mu_c \omega} \partial_\xi \vec{E}_c, \quad \partial_\xi \vec{E}_c \simeq -\frac{1}{\sigma} \partial_\xi^2 (\hat{n} \times \vec{H}_c),$$

so that

$$\partial_\xi^2 (\hat{n} \times \vec{H}_c) + \frac{2i}{\delta^2} (\hat{n} \times \vec{H}_c) = 0, \quad \delta := \sqrt{\frac{2}{\mu_c \omega \sigma}},$$

where δ is the **skin depth**. The solutions are

$$\vec{H}_c = \vec{H}_\parallel e^{-\xi/\delta} e^{i\xi/\delta}, \quad \vec{E}_c = \sqrt{\frac{\mu_c \omega}{2\sigma}} (1-i) (\hat{n} \times \vec{H}_\parallel) e^{-\xi/\delta} e^{i\xi/\delta}.$$

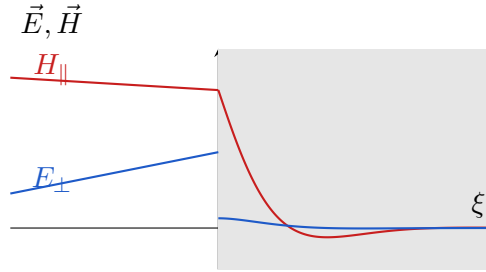


Figure 8.2: Qualitative behaviour of the fields near the surface of a good conductor (shaded): the tangential $\vec{H}$ is continuous and decays within a few skin depths, with oscillating phase.

The time-averaged power absorbed per unit area is

$$\frac{dP_{\text{loss}}}{da} = -\frac{1}{2} \text{Re}[\hat{n} \cdot \vec{E} \times \vec{H}^*] = \frac{\mu_c \omega \delta}{4} |\vec{H}_\parallel|^2.$$

With the effective surface current $\vec{K}_{\text{eff}} = \int_0^\infty d\xi \vec{J} = \hat{n} \times \vec{H}_\parallel$,

$$\frac{dP_{\text{loss}}}{da} = \frac{1}{2\sigma\delta} |\vec{K}_{\text{eff}}|^2,$$

where $1/\sigma\delta$ plays the role of a surface resistance.

8.2 Cylindrical Cavities and Waveguides

Assuming a sinusoidal time dependence $e^{-i\omega t}$ and no free sources,

$$\nabla \times \vec{E} = i\omega \vec{B}, \quad \nabla \cdot \vec{B} = 0, \quad \nabla \times \vec{B} = -i\mu\epsilon\omega \vec{E}, \quad \nabla \cdot \vec{E} = 0,$$

so

$$(\Delta + \mu\epsilon\omega^2) \begin{Bmatrix} \vec{E} \\ \vec{B} \end{Bmatrix} = 0.$$

Because of the cylindrical geometry we look for

$$\vec{E}(x, y, z, t) = \vec{E}(x, y) e^{\pm ikz - i\omega t}, \quad \vec{B}(x, y, z, t) = \vec{B}(x, y) e^{\pm ikz - i\omega t},$$

which gives

$$[\Delta_t + (\mu\varepsilon\omega^2 - k^2)] \begin{Bmatrix} \vec{E} \\ \vec{B} \end{Bmatrix} = 0,$$

where $\Delta_t = \Delta - \partial_z^2$ is the transverse part of the Laplacian. Decomposing the fields as $\vec{E} = E_z \hat{z} + \vec{E}_t$ (and similarly for $\vec{B}$),

$$\begin{aligned} \partial_z \vec{E}_t + i\omega \hat{z} \times \vec{B}_t &= \nabla_t E_z, & \hat{z} \cdot (\nabla_t \times \vec{E}_t) &= i\omega B_z, \\ \partial_z \vec{B}_t - i\mu\varepsilon\omega \hat{z} \times \vec{E}_t &= \nabla_t B_z, & \hat{z} \cdot (\nabla_t \times \vec{B}_t) &= -i\mu\varepsilon\omega E_z, \\ \nabla_t \cdot \vec{E}_t &= -\partial_z E_z, & \nabla_t \cdot \vec{B}_t &= -\partial_z B_z, \end{aligned}$$

and therefore (for e^{+ikz})

$$\begin{aligned} \vec{E}_t &= \frac{i}{\mu\varepsilon\omega^2 - k^2} [k \nabla_t E_z - \omega \hat{z} \times \nabla_t B_z], \\ \vec{B}_t &= \frac{i}{\mu\varepsilon\omega^2 - k^2} [k \nabla_t B_z + \mu\varepsilon\omega \hat{z} \times \nabla_t E_z]. \end{aligned}$$

The boundary conditions on a perfect conductor, $\hat{n} \times \vec{E} = 0$ and $\hat{n} \cdot \vec{B} = 0$, become

$$E_z|_S = 0, \quad \partial_n B_z|_S = 0.$$

TEM waves ($E_z = B_z = 0$) do not exist in a hollow cylinder. The two families of modes are

- **TM** (transverse magnetic): $B_z = 0$ everywhere, $E_z|_S = 0$;
- **TE** (transverse electric): $E_z = 0$ everywhere, $\partial_n B_z|_S = 0$.

8.3 Waveguides

The transverse fields are related by

$$\vec{H}_t = \pm \frac{1}{Z} \hat{z} \times \vec{E}_t,$$

where the $\pm$ comes from the dependence $e^{\pm ikz}$ and Z is the **wave impedance**

$$Z := \begin{cases} \frac{k}{\varepsilon\omega} = \frac{k}{k_0} \sqrt{\frac{\mu}{\varepsilon}}, & \text{TM,} \\ \frac{\mu\omega}{k} = \frac{k_0}{k} \sqrt{\frac{\mu}{\varepsilon}}, & \text{TE,} \end{cases} \quad k_0 = \omega \sqrt{\mu\varepsilon}.$$

With $\psi e^{\pm ikz}$ equal to E_z (TM) or H_z (TE),

$$\text{TM waves: } \vec{E}_t = \pm \frac{ik}{\gamma^2} \nabla_t \psi, \quad \text{TE waves: } \vec{H}_t = \pm \frac{ik}{\gamma^2} \nabla_t \psi,$$

where ψ satisfies

$$(\Delta_t + \gamma^2)\psi = 0, \quad \gamma^2 := \mu\varepsilon\omega^2 - k^2,$$

with $\psi|_S = 0$ (TM) or $\partial_n \psi|_S = 0$ (TE). For $\gamma^2 > 0$, ψ is oscillatory, and there is a discrete set of eigenvalues γ_λ^2 . Each mode has a **cutoff frequency**

$$\omega_\lambda := \frac{\gamma_\lambda}{\sqrt{\mu\varepsilon}}, \quad k_\lambda = \sqrt{\mu\varepsilon} \sqrt{\omega^2 - \omega_\lambda^2},$$

and phase velocity

$$v_p = \frac{\omega}{k_\lambda} = \frac{1}{\sqrt{\mu\varepsilon}} \frac{1}{\sqrt{1 - (\omega_\lambda/\omega)^2}}.$$

Modes with $\omega < \omega_\lambda$ cannot propagate; they are called **cutoff** or **evanescent modes**.

11 Special Theory of Relativity

11.1 The Situation before 1900, Einstein's Two Postulates

- Newton's second law is invariant under Galilean transformations,

$$\vec{x}' = \vec{x} - \vec{v}t, \quad t' = t.$$

- Maxwell's equations are not.

Einstein's postulates.

1. The laws of nature and the results of all experiments performed in a given frame of reference are independent of the translational motion of the system as a whole.
2. The speed of light is finite and independent of the motion of its source.
- 2'. In every inertial frame there is a finite universal limiting speed c for physical entities.

11.3 Lorentz Transformations and Basic Kinematic Results of Special Relativity

A. Simple Lorentz transformation of coordinates. The coordinates in K' , moving with velocity $v\hat{x}$ with respect to K , are related to those in K by

$$x'_0 = \gamma(x_0 - \beta x_1), \quad x'_1 = \gamma(x_1 - \beta x_0), \quad x'_2 = x_2, \quad x'_3 = x_3,$$

with $x_0 = ct$ and

$$\vec{\beta} := \frac{\vec{v}}{c}, \quad \gamma := \frac{1}{\sqrt{1 - \beta^2}}.$$

In general, for K' moving with velocity $\vec{v}$ with respect to K ,

$$x'_0 = \gamma(x_0 - \vec{\beta} \cdot \vec{x}), \quad \vec{x}' = \vec{x} + \frac{\gamma - 1}{\beta^2}(\vec{\beta} \cdot \vec{x})\vec{\beta} - \gamma\vec{\beta}x_0.$$

Another parametrisation uses the **rapidity** ζ :

$$\beta = \tanh \zeta, \quad \gamma = \cosh \zeta, \quad \gamma\beta = \sinh \zeta, \quad \gamma \in [1, \infty), \beta \in [0, 1).$$

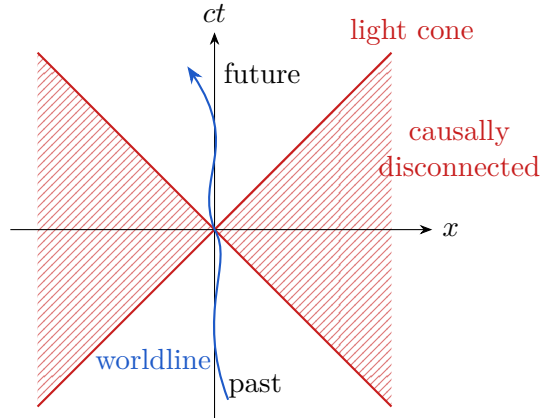


Figure 11.1: Light cone of an event at the origin and a timelike worldline through it.

C. Light cone, proper time, and time dilation. The invariant interval between two events is

$$s_{12}^2 = c^2(t_1 - t_2)^2 - |\vec{x}_1 - \vec{x}_2|^2, \quad \begin{cases} s_{12}^2 > 0, & \text{timelike (causally connected),} \\ s_{12}^2 < 0, & \text{spacelike (causally disconnected),} \\ s_{12}^2 = 0, & \text{lightlike.} \end{cases}$$

- If $s_{12}^2 > 0$, it is possible to find a reference frame in which $\vec{x}_1 = \vec{x}_2$.
- If $s_{12}^2 < 0$, it is possible to find a reference frame in which $t_1 = t_2$.
- If $s_{12}^2 = 0$, the events can be connected only by light signals.

Consider a particle moving with velocity $\vec{u}$ with respect to an inertial frame. The invariant

$$ds^2 = c^2 dt^2 - d\vec{x}^2 = c^2 dt^2 (1 - \beta^2) = \frac{c^2}{\gamma^2} dt^2$$

in the inertial frame equals, in the frame moving with the particle, $ds^2 = c^2 d\tau^2$, where τ is the **proper time**. Hence $d\tau = dt/\gamma$ and

$$\Delta t = \int_{\tau_1}^{\tau_2} \gamma(\tau) d\tau,$$

which is **time dilation** ($\gamma > 1$): a moving clock runs slower than a stationary one.

D. Relativistic Doppler shift. The phase of a wave is an invariant quantity (it counts wave crests in a wave train):

$$\phi = \omega t - \vec{k} \cdot \vec{x} = \omega' t' - \vec{k}' \cdot \vec{x}'.$$

Hence $(k_0, \vec{k})$, with $k_0 = \omega/c$, transforms like $(x_0, \vec{x})$:

$$k'_0 = \gamma(k_0 - \vec{\beta} \cdot \vec{k}), \quad k'_{\parallel} = \gamma(k_{\parallel} - \beta k_0), \quad \vec{k}'_{\perp} = \vec{k}_{\perp},$$

and, with $\hat{k} \cdot \hat{v} = \cos \theta$ and $\hat{k}' \cdot \hat{v} = \cos \theta'$,

$$\omega' = \gamma\omega(1 - \beta \cos \theta), \quad \tan \theta' = \frac{\sin \theta}{\gamma(\cos \theta - \beta)}.$$

11.4 Addition of Velocities; 4-Velocity

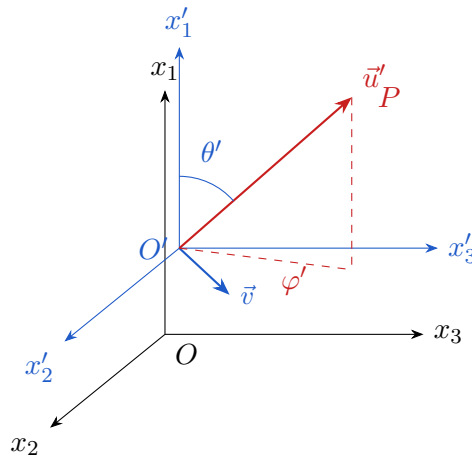


Figure 11.2: Frame K' moving with velocity $\vec{v}$ (along x_3) relative to K , and a velocity $\vec{u}'$ measured in K' .

For a boost along the direction of $\vec{v}$ (here the parallel axis), the inverse transformation of the differentials is

$$dx_0 = \gamma_v(dx'_0 + \beta dx'_\parallel), \quad dx_\parallel = \gamma_v(dx'_\parallel + \beta dx'_0), \quad d\vec{x}_\perp = d\vec{x}'_\perp,$$

which gives the **addition of velocities**:

$$u_\parallel = \frac{u'_\parallel + v}{1 + \frac{\vec{v} \cdot \vec{u}'}{c^2}}, \quad \vec{u}_\perp = \frac{\vec{u}'_\perp}{\gamma_v \left(1 + \frac{\vec{v} \cdot \vec{u}'}{c^2}\right)},$$

or, in terms of magnitudes and angles,

$$\tan \theta = \frac{u' \sin \theta'}{\gamma_v(u' \cos \theta' + v)}, \quad u = \frac{\sqrt{u'^2 + v^2 + 2u'v \cos \theta' - \left(\frac{u'v \sin \theta'}{c}\right)^2}}{1 + \frac{u'v}{c^2} \cos \theta'}.$$

The addition law is not a Lorentz transformation, because time is not invariant under such transformations. However, proper time is invariant, so we define the **4-velocity**

$$U_0 := \frac{dx_0}{d\tau} = \frac{dx_0}{dt} \frac{dt}{d\tau} = \gamma_u c, \quad \vec{U} := \frac{d\vec{x}}{d\tau} = \frac{d\vec{x}}{dt} \frac{dt}{d\tau} = \gamma_u \vec{u},$$

which is related to the total energy and momentum of a particle.

11.5 Relativistic Momentum and Energy of a Particle

The total energy of a particle of mass m and velocity $\vec{u}$ is

$$E = \gamma m c^2 = \frac{m c^2}{\sqrt{1 - u^2/c^2}}.$$

The invariant “length” of the energy–momentum 4-vector ($p_0 = E/c$, $\vec{p}$) is

$$p_0^2 - \vec{p}^2 = (mc)^2 \implies E = \sqrt{c^2 p^2 + m^2 c^4}.$$

11.10 Transformation of Electromagnetic Fields

From here on the notes use Gaussian units, as in the corresponding chapters of Jackson.

The field-strength tensor is

$$F^{\alpha\beta} := \partial^\alpha A^\beta - \partial^\beta A^\alpha,$$

which transforms as a second-rank tensor,

$$F'^{\alpha\beta} = \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu} F^{\mu\nu}.$$

This gives

$$\begin{aligned} \vec{E}' &= \gamma(\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{E}), \\ \vec{B}' &= \gamma(\vec{B} - \vec{\beta} \times \vec{E}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{B}). \end{aligned}$$

In SI units ($\vec{B} \rightarrow c\vec{B}$), and using $\frac{\gamma^2}{\gamma+1}\beta^2 = \gamma - 1$,

$$\begin{aligned} \vec{E}' &= \gamma(\vec{E} + c\vec{\beta} \times \vec{B}) - (\gamma - 1) \hat{\beta}(\hat{\beta} \cdot \vec{E}), \\ \vec{B}' &= \gamma\left(\vec{B} - \frac{\vec{\beta} \times \vec{E}}{c}\right) - (\gamma - 1) \hat{\beta}(\hat{\beta} \cdot \vec{B}). \end{aligned}$$

13 Collisions, Energy Loss, and Scattering of Charged Particles; Cherenkov and Transition Radiation

This chapter has a title page only in the handwritten notes.

14 Radiation by Moving Charges

14.1 Liénard–Wiechert Potentials and Fields for a Point Charge

The 4-potential of a charged particle in motion is (Gaussian units)

$$A^\alpha(x) = \frac{4\pi}{c} \int d^4x' D_r(x - x') J^\alpha(x'),$$

where x is the observation point, x' the source point, and

$$D_r(x - x') = \frac{1}{2\pi} \theta(x_0 - x'_0) \delta[(x - x')^2]$$

is the retarded Green function. For a point charge with 4-velocity $V^\alpha(\tau)$ and position $r(\tau)$,

$$J^\alpha(x') = ec \int d\tau V^\alpha(\tau) \delta^{(4)}(x' - r(\tau)),$$

so that

$$A^\alpha(x) = 2e \int d\tau V^\alpha(\tau) \underbrace{\theta(x_0 - r_0(\tau))}_{\text{retardation}} \underbrace{\delta([x - r(\tau)]^2)}_{\text{light-cone condition}}.$$

Using

$$\delta[f(x)] = \sum_i \frac{\delta(x - x_i)}{|\left(\frac{df}{dx}\right)_{x=x_i}|}, \quad \frac{d}{d\tau}[x - r(\tau)]^2 = -2[x - r(\tau)]_\beta V^\beta(\tau),$$

we obtain the **Liénard–Wiechert potentials**

$$A^\alpha(x) = \frac{e V^\alpha(\tau)}{V \cdot [x - r(\tau)]} \Big|_{\tau=\tau_0},$$

where τ_0 is fixed by the light-cone constraint $x_0 - r_0(\tau_0) = |\vec{x} - \vec{r}(\tau_0)| := R$. With $\hat{n} = (\vec{x} - \vec{r})/R$,

$$V \cdot [x - r(\tau)] = V_0 [x_0 - r_0(\tau_0)] - \vec{V} \cdot [\vec{x} - \vec{r}(\tau_0)] = \gamma c R - \gamma \vec{v} \cdot \hat{n} R = \gamma c R (1 - \vec{\beta} \cdot \hat{n}),$$

and therefore

$$\phi(\vec{x}, t) = \left[\frac{e}{R(1 - \vec{\beta} \cdot \hat{n})} \right]_{\text{ret}}, \quad \vec{A}(\vec{x}, t) = \left[\frac{e \vec{\beta}}{R(1 - \vec{\beta} \cdot \hat{n})} \right]_{\text{ret}},$$

where “ret” means evaluated at $r_0(\tau_0) = x_0 - R$.

For later use, with $x^0 = ct$ and $\gamma d\tau = dt$,

$$U^0 = \frac{dx^0}{d\tau} = \frac{dx^0}{dt} \frac{dt}{d\tau} = c\gamma, \quad \frac{dU^0}{d\tau} = c \frac{d\gamma}{d\tau} = c\gamma^4 \vec{\beta} \cdot \dot{\vec{\beta}},$$

using $d\gamma/d\beta = \gamma^3 \beta$ and $\dot{\vec{\beta}} = d\vec{\beta}/dt$.

In covariant form, with $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$,

$$F^{\alpha\beta} = \frac{e}{V \cdot (x - r)} \frac{d}{d\tau} \left[\frac{(x - r)^\alpha V^\beta - (x - r)^\beta V^\alpha}{V \cdot (x - r)} \right].$$

In non-covariant form,

$$\begin{aligned} \vec{B} &= [\hat{n} \times \vec{E}]_{\text{ret}}, \\ \vec{E} &= e \underbrace{\left[\frac{\hat{n} - \vec{\beta}}{\gamma^2(1 - \vec{\beta} \cdot \hat{n})^3 R^2} \right]_{\text{ret}}}_{\text{“velocity field”} \sim 1/R^2} + \frac{e}{c} \underbrace{\left[\frac{\hat{n} \times \{(\hat{n} - \vec{\beta}) \times \dot{\vec{\beta}}\}}{(1 - \vec{\beta} \cdot \hat{n})^3 R} \right]_{\text{ret}}}_{\text{“acceleration field”} \sim 1/R}. \end{aligned}$$

14.2 Total Power Radiated by an Accelerated Charge: Larmor’s Formula and its Relativistic Generalisation

If $\dot{\vec{\beta}} \neq 0$ and $\beta \ll 1$, the acceleration field is

$$\vec{E}_a \simeq \frac{e}{c} \left[\frac{\hat{n} \times (\hat{n} \times \dot{\vec{\beta}})}{R} \right]_{\text{ret}},$$

and the Poynting vector is

$$\vec{S} = \frac{c}{4\pi} \vec{E} \times \vec{B} = \frac{c}{4\pi} |\vec{E}_a|^2 \hat{n}.$$

The power radiated per unit solid angle is then

$$\frac{dP}{d\Omega} = \frac{c}{4\pi} |R\vec{E}_a|^2 = \frac{e^2}{4\pi c} \left| \hat{n} \times (\hat{n} \times \dot{\vec{\beta}}) \right|^2 = \frac{e^2}{4\pi c^3} |\dot{\vec{v}}|^2 \sin^2 \Theta,$$

where Θ is the angle between $\hat{n}$ and the acceleration $\dot{\vec{v}}$. Integrating over angles gives the **Larmor formula** for non-relativistic particles:

$$P = \frac{2}{3} \frac{e^2}{c^3} |\dot{\vec{v}}|^2.$$

The Lorentz-invariant generalisation is

$$P = -\frac{2}{3} \frac{e^2}{m^2 c^3} \frac{dp_\mu}{d\tau} \frac{dp^\mu}{d\tau},$$

where, with $E = \gamma m c^2$ and $\vec{p} = \gamma m \vec{v}$,

$$-\frac{dp_\mu}{d\tau} \frac{dp^\mu}{d\tau} = \left(\frac{d\vec{p}}{d\tau} \right)^2 - \frac{1}{c^2} \left(\frac{dE}{d\tau} \right)^2 = \left(\frac{d\vec{p}}{d\tau} \right)^2 - \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2.$$

This gives the **Liénard result** (1898)

$$P = \frac{2}{3} \frac{e^2}{c} \gamma^6 \left[\dot{\vec{\beta}}^2 - (\vec{\beta} \times \dot{\vec{\beta}})^2 \right].$$